Supplementary Material for: CFINet: A Coarse–Fine Interaction Network for Gross Tumor Volume Segmentation in Esophageal Cancer Radiotherapy

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Abstract

Gross tumor volume (GTV) delineation is a critical step in radiotherapy planning for esophageal cancer. Manual contouring on radiotherapy computed tomography (RTCT) images is time-consuming and subject to inter-observer variability. Automatic segmentation remains challenging due to variations in tumor location and extent and weak contrast with surrounding tissues, requiring effective integration of global contextual information and fine-grained structural details. To address these challenges, we propose CFINet, a coarse–fine interaction transformer architecture for esophageal GTV segmentation. The framework combines a multi-scale branch for fine-grained structural representation with a fixed-resolution branch for coarse contextual modeling. Grouped self-attention and grouped cross-attention facilitate efficient feature modeling and interaction within and between the branches, while a cross-scale decoder integrates complementary features for segmentation. CFINet was evaluated on an internal GTV dataset and a temporally held-out same-institution test cohort. The public SegTHOR dataset was separately used to assess thoracic organs-at-risk segmentation. CFINet achieved Dice similarity coefficients (DSCs) of 0.7424 ± 0.1442 and 0.7116 ± 0.1349 on the two GTV cohorts, respectively, and a mean DSC of 0.8833 ± 0.0466 on SegTHOR. These results support the effectiveness of coarse–fine feature integration for esophageal GTV segmentation, with complementary evidence from thoracic organ segmentation. CFINet shows potential to provide initial contours for physician review and refinement in radiotherapy planning.

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1 Experimental Setup

1.1 Datasets and Preprocessing

Dataset A comprised 38 planning CT scans collected retrospectively between June 2020 and September 2022. Dataset B included eight subsequently collected cases from June 2023 to April 2024. The two cohorts originated from the same institution and were acquired using the same scanner and imaging protocol. Dataset A was used for patient-level five-fold evaluation and for training the final models, while Dataset B was reserved for temporally held-out, same-institution testing. No Dataset B images or reference contours were used for model development or adaptation.

The reference GTV contours were extracted from radiotherapy planning RTStruct files. Each GTV was initially delineated by an attending radiation oncologist and subsequently reviewed by a senior radiation oncologist with over 30 years of clinical experience, holding the rank of chief physician. When disagreements arose, the two physicians discussed the affected regions, and the contours were corrected where necessary. The final reference contours were confirmed by the chief physician. Delineation was informed by available clinical information, including endoscopy, esophagography, MRI, and PET findings when available. The same delineation and review procedure was followed for both cohorts. The study was approved by the Biological and Medical Ethics Committee of Northeastern University (Approval No. NEU-EC-2021B016S, March 17, 2021), which waived the requirement for informed consent owing to the retrospective study design.

The public SegTHOR dataset [10] was evaluated separately as a thoracic organ-at-risk segmentation benchmark, comprising the heart, aorta, trachea, and esophagus. It was not treated as an independent-site test cohort for esophageal GTV segmentation. Additional SegTHOR results are provided in Section 6.

Table 1 Preprocessing and augmentation settings of CFINet for esophageal GTV segmentation. Operations marked as not applied at inference were used only during training.

Operation	Training	Inference
Image loading	Image and reference label	Image and reference label loaded for evaluation (ITKReader)
Orientation	RAS for image and label	RAS for the input image
Voxel spacing	$1.87 \times 1.87 \times 5.0\,\mathrm{mm}$ (bilinear image and nearest-neighbor label resampling)	Same spacing with image-only resampling
Intensity normalization	HU clipping $[-175, 250]$, mapped to $[0, 1]$	Same clipping and scaling
Patch sampling / inference	$128 \times 128 \times 64$ voxels, positive/negative weights 1:1, two samples per call	Sliding-window inference with full-volume reconstruction
Random flipping	Each spatial axis independently ($p = 0.3$)	Not applied
Random 90° rotation	First two spatial axes ($p = 0.3$, $k \in \{1, 2, 3\}$)	Not applied
Intensity scaling	Random factor in $[-0.10, 0.10]$ ($p = 0.3$)	Not applied
Intensity shifting	Random offset in $[-0.10, 0.10]$ ($p = 0.3$)	Not applied

Table 1 summarizes the preprocessing and augmentation settings used by CFINet for the esophageal GTV experiments. CT images were reoriented to RAS and resampled to a voxel spacing of $1.87 \times 1.87 \times 5.0\,\mathrm{mm}$. Intensities were clipped to $[-175, 250]$ HU and mapped linearly to $[0, 1]$ according to

$$I_{\mathrm{norm}} = \frac{\mathrm{clip}(I, -175, 250) + 175}{425}. \tag{1}$$

Training involved label-guided random patch sampling and spatial and intensity augmentation. At inference, CT volumes were resampled and intensity-normalized before sliding-window inference, without random augmentation. The window-wise predictions were aggregated into a full-volume segmentation and restored to the original CT image space for evaluation against the reference GTV annotations.

1.2 Five-Fold Evaluation

Dataset A was evaluated using patient-level five-fold train–test cross-validation. In each fold, 30–31 patients were used for training, and the remaining 7–8 patients were held out for testing. No separate validation set was used. CFINet was optimized using AdamW [13] and a combined Dice and cross-entropy loss. The learning rate followed a linear-warmup cosine-annealing schedule (`LinearWarmupCosineAnnealingLR`), increasing to 1×10^{-4} over the first 1000 epochs before gradually decreasing toward $\eta_{\min} = 1 \times 10^{-6}$ by epoch 4000. All five folds were trained for 4000 epochs using the same optimization settings. The final checkpoint from each fold was used to evaluate the corresponding held-out patients, without early stopping or checkpoint selection based on test performance.

Predictions from the five held-out folds were combined into a single out-of-fold evaluation set, with each patient contributing exactly one prediction. All cohort-level summaries and patient-level comparisons for Dataset A were computed from these held-out predictions rather than from training predictions or an unweighted average of fold-level summary scores.

For reproducibility, the random seed was set to 1 for NumPy and PyTorch, including CUDA random-number generation. The cuDNN deterministic setting was enabled (`torch.backends.cudnn.deterministic=True`), and cuDNN benchmarking was disabled (`torch.backends.cudnn.benchmark=False`). These settings were used to reduce run-to-run variation without assuming that every operation was fully deterministic.

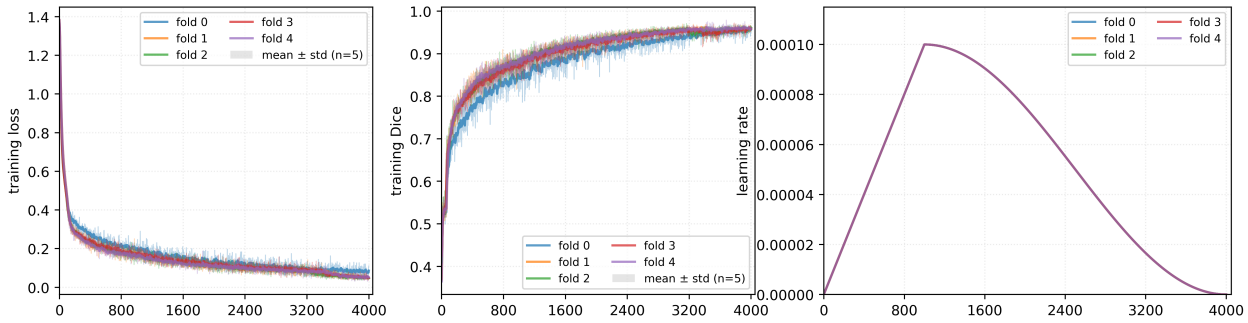

Figure 1 Training dynamics of CFINet across the five patient-level folds of Dataset A: (a) training loss, (b) training DSC, and (c) learning rate. Individual fold curves are shown. Shaded bands in (a) and (b) represent the mean ± standard deviation across folds. The learning rate increases to 1×10^{-4} during the first 1000 epochs and subsequently decreases toward $\eta_{\min} = 1 \times 10^{-6}$ at epoch 4000. Training DSC includes both background and foreground classes, whereas test DSC evaluates the foreground GTV only. These values are not directly comparable. The final checkpoint was used in each fold without early stopping or checkpoint selection based on held-out test performance.

Fig. 1 presents the training loss, training DSC, and learning-rate curves across the five folds. Training loss decreases steadily, while training DSC rises and changes more gradually during later epochs. The learning-rate curves follow the specified 1000-epoch warm-up and cosine annealing to 1×10^{-6}. Training DSC includes both background and foreground classes and is measured on sampled patches. In contrast, the reported test DSC is calculated for the foreground GTV from predictions restored to the original CT space. The two DSC measurements therefore have different definitions and should not be compared directly. The training curves were used to document optimization rather than to choose an evaluation checkpoint.

1.3 Temporal Testing

For temporal testing, each method was retrained on the full Dataset A cohort using its model-specific configuration. The final checkpoint was then used to evaluate the eight cases in Dataset B, without selecting a particular cross-validation fold or combining multiple fold-specific models. Dataset B was used only for testing and was not involved in model training, hyperparameter selection, or fine-tuning.

1.4 Baselines and Metrics

Eighteen baseline methods were included to compare CFINet with convolutional, Transformer-based, and hybrid segmentation architectures. Table 2 summarizes their input sizes, optimizers, initial learning rates, training durations, learning-rate schedules, and pretraining settings. All methods used the same patient-level partitions of Dataset A and were evaluated on the same temporally held-out Dataset B cohort. The patient assignments and reference GTV annotations were consistent across methods, while preprocessing, augmentation, and optimization followed each implementation.

For the framework-based comparisons, nnUNet [7] was implemented using the nnUNet v2 two-stage 3D cascade, consisting of `3d_lowres` and `3d_cascade_fullres`. Both stages were trained for 1000 epochs in each of the five folds. The generated plans specified patch sizes of $192 \times 160 \times 64$ and $192 \times 192 \times 48$ (H×W×D) at voxel spacings of $1.871 \times 1.871 \times 5.0$ mm and $0.977 \times 0.977 \times 5.0$ mm, respectively. The cascade full-resolution stage inherited the configuration of `3d_fullres`. The standard nnUNet v2 preprocessing and optimization settings were used, with patient-level training and test assignments specified to match the study partitions. Final checkpoints at the designated training endpoints were used for evaluation. PHTrans [12] and TransHRNet [21] used their respective official implementations based on nnUNet v1, with the same patient-level partitions.

The remaining methods were trained with their model-specific configurations listed in Table 2. The duration was specified in epochs or optimizer iterations according to each training implementation. TransBTS [18] and 3D UX-Net [11] shared an iteration-based training framework. TransUNet [2] and SwinUNet [1] used ImageNet-pretrained backbones. The other baselines were trained without pretrained weights. The reported five-fold comparisons used the final checkpoints from the corresponding training runs, rather than selecting intermediate checkpoints based on held-out test DSC.

Table 2 Model-specific training configurations of the eighteen baseline methods. Three-dimensional patch dimensions are reported as height × width × depth (H×W×D). Two-dimensional inputs are reported as height × width. Training duration is reported in epochs or optimizer iterations according to the corresponding implementation. Only TransUNet and SwinUNet used pretrained weights.

Model	Input size	Optimizer / initial LR	Training epochs / iterations	LR schedule	Pretrain
VNet [14]	$128 \times 128 \times 64$	AdamW / 10^{-4}	1000 epochs	Warm-up cosine	No
AttentionUNet [15]	$128 \times 128 \times 64$	AdamW / 10^{-4}	1000 epochs	Warm-up cosine	No
UNet++ [24]	$128 \times 128 \times 64$	AdamW / 10^{-4}	1000 epochs	Warm-up cosine	No
DenseUNet [22]	$128 \times 128 \times 32$	AdamW / 10^{-4}	4000 epochs	Warm-up cosine	No
PSNN [8]	$128 \times 128 \times 32$	AdamW / 10^{-4}	1000 epochs	Warm-up cosine	No
nnUNet (v2 cascade) [7]	Low: $192 \times 160 \times 64$ (low resolution), $192 \times 192 \times 48$ (full resolution)	SGD / 10^{-2}	1000 epochs per stage	Polynomial	No
TransUNet [2]	2D: 224×224	SGD / 10^{-4}	1000 epochs	Polynomial	ImageNet
SwinUNet [1]	2D: 224×224	SGD / 10^{-4}	1000 epochs	Polynomial	ImageNet
TransBTS [18]	$96 \times 96 \times 96$	AdamW / 10^{-4}	50 000 iterations	Constant	No
3D UX-Net [11]	$128 \times 128 \times 32$	AdamW / 10^{-4}	50 000 iterations	Constant	No
nnFormer [23]	$128 \times 128 \times 64$	AdamW / 10^{-4}	1000 epochs	Warm-up cosine	No
UNETR [6]	$128 \times 128 \times 64$	AdamW / 10^{-4}	5000 epochs	Warm-up cosine	No
SwinUNETR [5]	$128 \times 128 \times 64$	AdamW / 10^{-4}	5000 epochs	Warm-up cosine	No
DiffUNet [20]	$96 \times 96 \times 96$	AdamW / 2×10^{-4}	2000 epochs	Warm-up cosine	No
PHTrans [12]	$192 \times 192 \times 48$	SGD / 10^{-2}	1000 epochs	Polynomial	No
TransHRNet [21]	$192 \times 192 \times 48$	SGD / 10^{-2}	1000 epochs	Polynomial	No
MedFormer [3]	$128 \times 128 \times 64$	AdamW / 10^{-5}	4000 epochs	Warm-up cosine	No
UNETR++ [16]	$128 \times 128 \times 64$	SGD / 10^{-2}	1000 epochs	Polynomial	No

For all methods, predictions were restored to the original CT space before evaluation against the reference GTV contours. Segmentation performance was assessed using foreground DSC, Jaccard, recall, and precision, together with HD95 and ASSD in millimeters. Model parameters and multiply–accumulate operations (MACs) were used to characterize model complexity, with MACs computed using THOP [25]. Inference time and peak GPU memory were measured under the evaluation settings described in this study. Additional patient-level statistical comparisons and performance–efficiency analyses are provided in Section 2.

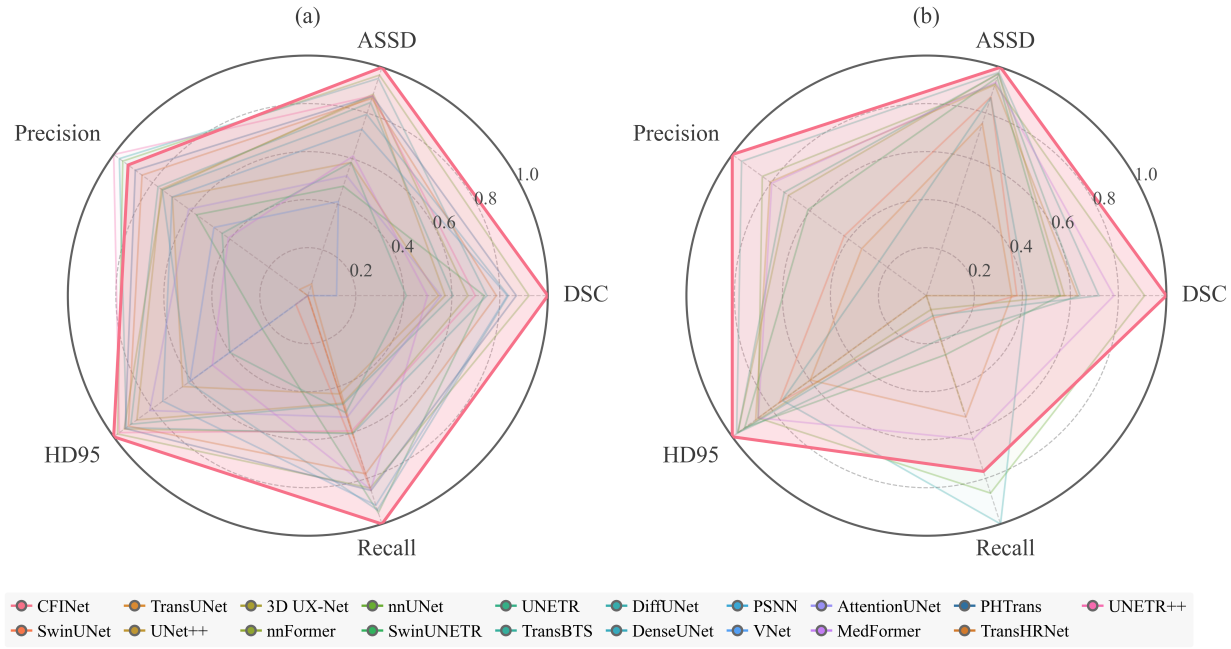

Figure 2 Radar-chart summary of normalized multi-metric segmentation performance. (a) Dataset A. (b) Public dataset results, discussed in Section 6. For ASSD and HD95, the values are direction-normalized so that larger radial extent indicates better performance.

2 Performance and Efficiency

2.1 Dataset A Results

The complete quantitative results on Dataset A are reported in the main manuscript and are not repeated here. To provide a compact multi-metric view, Fig. 2(a) presents a radar-chart summary of the Dataset A results across DSC, ASSD, Precision, HD95, and Recall. For ASSD and HD95, the values were direction-normalized so that a larger radial extent consistently indicates better performance. Fig. 2(b) shows the corresponding summary on the public dataset and is discussed in Section 6.

As shown in Fig. 2(a), CFINet shows a favorable profile across the normalized overlap and boundary metrics. The radar plot provides a visual summary of these metrics, while the quantitative comparisons remain the basis for judging individual measures. It achieved the highest mean DSC and recall and the lowest mean HD95 and ASSD on Dataset A. Several competing methods showed greater differences between overlap, precision, and surface-distance performance.

2.2 Paired Statistical Tests

To examine whether the mean performance differences were also reflected at the patient level, DSC was selected as the paired comparison endpoint. Six baseline methods were selected for patient-level paired statistical analysis on Dataset A, considering their segmentation performance and architectural relevance to CFINet. These included nnUNet (cascade) and DenseUNet as convolutional comparisons, PHTrans and TransHRNet for their parallel and multi-resolution feature representations, and MedFormer and UNETR++ for their compact representations and efficient attention designs.

We used two-sided Wilcoxon signed-rank tests for paired comparisons. We also calculated the mean patient-level DSC difference (CFINet minus each baseline) and its 95% confidence interval using 10,000 paired bootstrap resamples. A Friedman test was also performed across the seven methods included in this paired analysis.

Table 3 shows that CFINet achieved positive mean paired DSC differences against all six compared baselines. The mean paired difference was 0.0310 relative to nnUNet (cascade), with a 95% confidence interval of [0.0036, 0.0659], and 0.0519 relative to PHTrans, with a 95% confidence interval of [0.0014, 0.1036]. The corresponding Wilcoxon p values were 0.0336 and 0.0298. Larger paired advantages were observed relative to UNETR++, MedFormer, DenseUNet, and TransHRNet. The Friedman test also indicated an overall difference in patient-level DSC among the seven methods ($\chi^2(6) = 50.4192$, $p = 3.87 \times 10^{-9}$).

Table 3 Patient-level paired DSC comparison on Dataset A. Mean differences are calculated as CFINet minus the baseline method. Confidence intervals are estimated from 10,000 paired bootstrap resamples. Reported p values are two-sided Wilcoxon signed-rank test results. r_{rb} denotes the matched-pairs rank-biserial correlation.

Comparison	Mean DSC difference	95% CI	p value	r_{rb}
CFINet vs. nnUNet (cascade)	0.0310	[0.0036, 0.0659]	0.0336	0.3997
CFINet vs. DenseUNet	0.0651	[0.0126, 0.1187]	0.0096	0.4822
CFINet vs. UNETR++	0.1197	[0.0618, 0.2031]	0.0046	0.5249
CFINet vs. TransHRNet	0.0848	[0.0369, 0.1592]	0.0253	0.4196
CFINet vs. PHTrans	0.0519	[0.0014, 0.1036]	0.0298	0.4083
CFINet vs. MedFormer	0.1986	[0.1498, 0.2567]	< 0.001	0.9317

2.3 Computational Efficiency

To examine computational efficiency alongside segmentation performance, we compared CFINet with six representative baselines: nnUNet (cascade), DenseUNet, PHTrans, TransHRNet, MedFormer, and UNETR++. These methods include competitive baselines and architectures that explore parallel and multi-resolution feature representations or efficient attention mechanisms.

Table 4 reports the parameter counts, MACs, FP32 inference time, and peak GPU memory. For nnUNet (cascade), the complexity analysis considers both the low-resolution and cascade full-resolution networks, rather than profiling only a single stage. The inference time and peak GPU memory correspond to the complete two-stage inference pipeline.

CFINet achieved the highest DSC among the methods included in this efficiency comparison. It required 1517.82 G MACs and 56.62 M parameters, with a measured FP32 inference time of 2.98 s per CT volume and a peak GPU memory usage of 2.28 GiB. Compared with nnUNet (cascade), CFINet obtained a higher DSC while reducing inference time from 6.60 to 2.98 s per volume and reducing peak GPU memory from 2.63 to 2.28 GiB under the reported evaluation settings. PHTrans and TransHRNet were faster, and UNETR++ required fewer MACs and parameters. Each achieved a lower mean DSC. CFINet therefore offered higher segmentation accuracy at a higher computational cost than several baselines, although it was faster than the nnUNet cascade under the measured inference conditions.

Table 4 Segmentation performance and computational characteristics of CFINet and selected methods. Higher DSC is better. Lower values are preferred for Params, MACs, inference time, and peak GPU memory.

Model	DSC ↑	Params (M) ↓	MACs (G) ↓	Time (s/vol.) ↓	Peak memory (GiB) ↓
CFINet	0.7424	56.6203	1517.82	2.9759	2.2763
nnUNet (cascade)	0.7114	59.7161	628.06	6.6018	2.6261
PHTrans	0.6905	36.8755	445.69	1.0002	1.8002
DenseUNet	0.6773	49.1589	547.02	2.7990	1.7759
TransHRNet	0.6576	34.2103	569.29	0.5615	1.1098
UNETR++	0.6227	34.6413	63.74	0.9857	0.6917
MedFormer	0.5439	36.6670	209.11	3.4542	1.1727

2.4 Qualitative GTV Results

Figs. 4 and 5 show two representative cases from Dataset A, showing the GTV segmentation results of CFINet and all eighteen baseline methods. The sagittal reference images indicate the anatomical location of each tumor along the superior–inferior direction of the esophagus. The axial CT images provide an overview of the

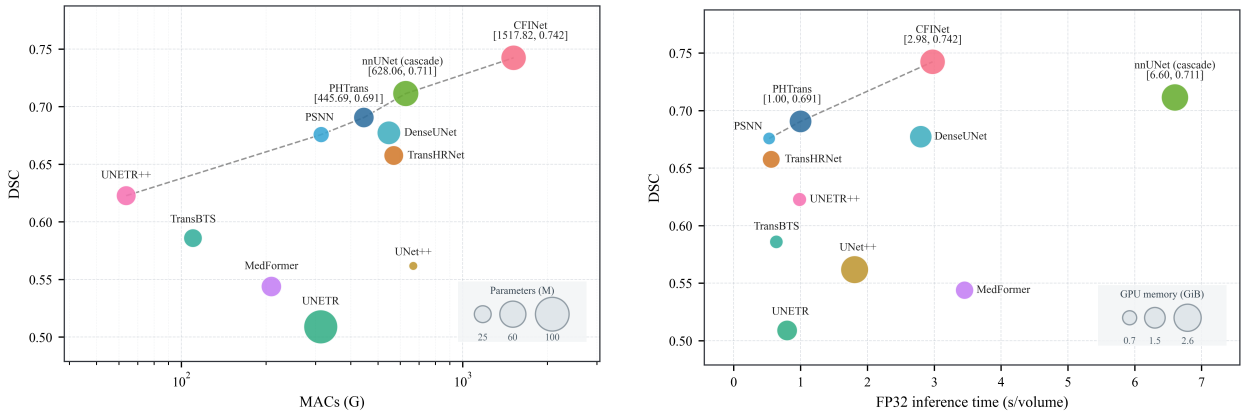

Figure 3 Performance–efficiency comparison of CFINet and representative methods. (a) DSC versus MACs, where bubble size indicates the number of parameters. (b) DSC versus FP32 inference time per CT volume, where bubble size indicates peak GPU memory usage.

target region, while the enlarged axial views allow a closer comparison between the predicted masks and the reference GTV contours. Slice-level DSC values are displayed above the predictions.

In the first case (Fig. 4), most methods identified the tumor region, although differences in the predicted contours can be observed in the enlarged views. For the displayed slice, CFINet achieved the highest DSC of 0.9116, followed closely by DiffUNet (0.9088). DenseUNet, UNETR++, SwinUNETR, and several other methods also produced segmentations with relatively high overlap. Compared with the reference contour, CFINet showed good agreement in the depicted region, whereas some methods exhibited local under-segmentation or deviations along the tumor boundary.

The second case (Fig. 5) shows greater variation among the compared methods. For the displayed slice, TransHRNet achieved the highest DSC of 0.8993, followed by CFINet (0.8949), DiffUNet (0.8929), and SwinUNETR (0.8911). Their predictions generally captured the target region, with differences mainly observed around the tumor boundaries. In contrast, several other methods showed substantial under-segmentation or failed to identify the target. These two examples provide a visual comparison of tumor localization and contour agreement, complementing the quantitative evaluation on Dataset A.

3 Ablation and ERF Analysis

3.1 Ablation Results

The Dataset A ablations evaluate grouped self-attention (GSA), grouped cross-attention (GCA), cross-scale feature fusion (CSFF), and the Path-2 patch size. Table 5 lists the model configurations and segmentation results. A0 uses a single hierarchical path with spatial-reduction attention (SRA), and A1 replaces SRA with GSA. A2 adds a fixed-resolution second path without cross-attention. A3 uses standard cross-attention (CA) between the paths, while A4 uses GCA. A4b uses CSFF only at Stage 4, without shallow Path-2 decoding. A5 uses multi-stage CSFF throughout the decoder. A5_patch8 changes only the Path-2 patch size from 16^3 to 8^3.

Replacing SRA with GSA improved DSC from 0.6160 to 0.6443 and recall from 0.5331 to 0.5652, although HD95 and ASSD increased. The boundary results are further examined in Section 3.2. Adding cross-attention to the dual-path model increased DSC from 0.6340 in A2 to 0.6559 in A3 and 0.6514 in A4. GCA produced a slightly lower DSC than standard CA while reducing attention-related computation, as detailed in Section 3.3.

The full A5 configuration achieved a DSC of 0.7424, compared with 0.6892 for A4b. Jaccard increased from 0.5548 to 0.6059 and recall from 0.5963 to 0.7267. Precision decreased from 0.8946 to 0.8117, while ASSD changed little (4.6877 versus 4.5909 mm). The higher recall indicates that A5 recovered a larger proportion of

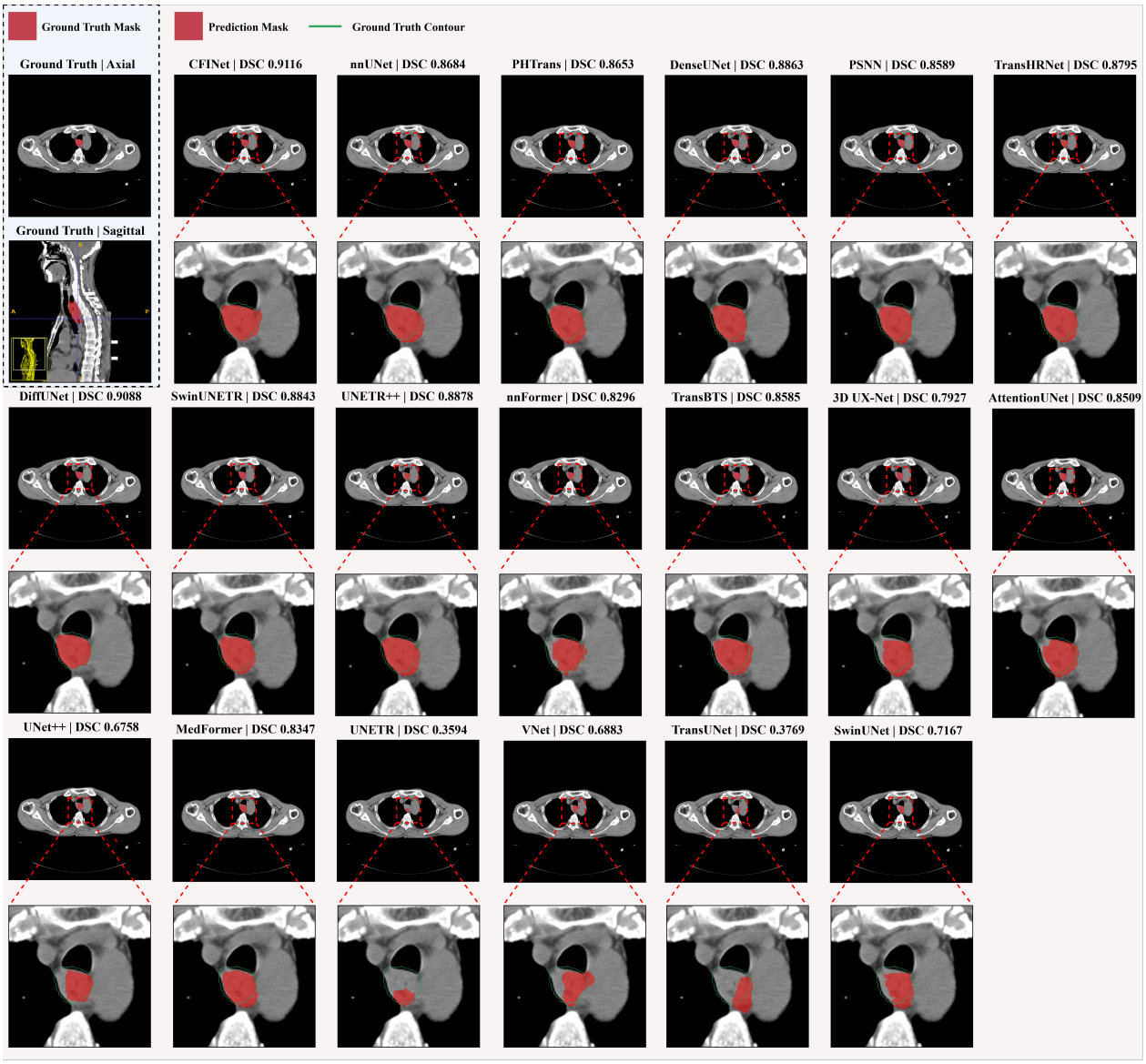

Figure 4 Visualization of GTV segmentation results for the first representative case from Dataset A. The leftmost column shows the reference axial and sagittal CT images. The sagittal view indicates the superior–inferior location of the GTV within the esophagus. For each method, the upper image shows the axial CT slice with the predicted GTV, and the lower image provides an enlarged axial view of the target region. Red denotes the segmentation mask, and green outlines indicate the reference GTV contour. The slice-level DSC is reported above each prediction.

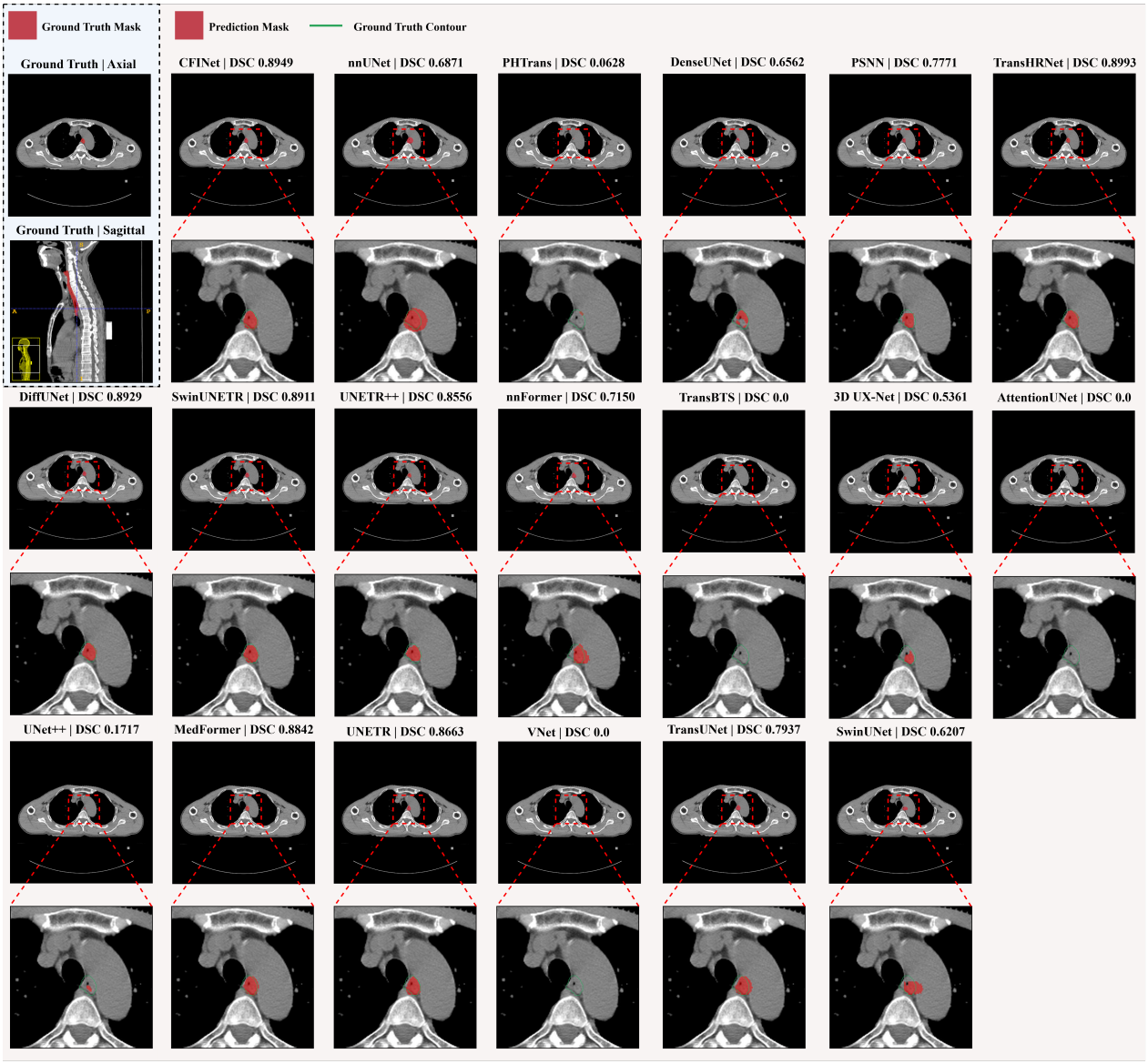

Figure 5 Visualization of GTV segmentation results for the second representative case from Dataset A, using the same layout and color scheme as Fig. 4. The sagittal reference image provides anatomical context for tumor location, while the enlarged axial views illustrate differences in target delineation among the compared methods. Slice-level DSC values are shown above the corresponding predictions.

Table 5 Ablation results on Dataset A. Red , blue , and green indicate the best, second-best, and third-best values, respectively. HD95 and ASSD are reported in millimeters.

Variant	Model configuration	DSC	Jaccard	HD95	Precision	Recall	ASSD
A0	Single-path baseline with SRA	0.6160	0.4696	27.1199	0.8287	0.5331	7.3421
A1	Single-path baseline with GSA	0.6443	0.4949	67.2085	0.8334	0.5652	11.0707
A2	Dual-path GSA without cross-attention	0.6340	0.4770	23.1051	0.7870	0.5634	6.9292
A3	Dual-path GSA with standard cross-attention	0.6559	0.5077	39.0993	0.7170	0.6443	7.1075
A4	Dual-path GSA with GCA, without CSFF	0.6514	0.5110	40.8102	0.8417	0.5893	6.1243
A4b	Dual-path GSA + GCA, Stage-4 CSFF	0.6892	0.5548	29.3972	**0.8946**	0.5963	4.6877
A5 (Ours)	Full CFINet with Path-2 patch size 16^3	**0.7424**	**0.6059**	21.6010	0.8117	**0.7267**	**4.5909**
A5_patch8	Full CFINet with Path-2 patch size 8^3	0.7089	0.5618	**20.2365**	0.7953	0.6655	5.3485

A5 uses GSA, GCA, and multi-stage CSFF. A4b uses Stage-4 CSFF without shallow Path-2 decoding. A5_patch8 differs from A5 only in the Path-2 patch size (8^3 vs. 16^3). SRA denotes spatial-reduction attention.

annotated tumor voxels, consistent with its higher overlap. With smaller Path-2 patches, A5_patch8 achieved a lower DSC (0.7089) and recall (0.6655), although its HD95 was slightly lower (20.24 versus 21.60 mm). The 16^3 patch size was therefore retained in CFINet.

Table 6 compares the computational requirements of these configurations. From A0 to A1, the parameter count decreased from 42.60 to 29.29 M and inference time from 1.70 to 1.52 s per volume, with little change in MACs. In this single-path comparison, GSA improved DSC without increasing the measured computational cost.

Table 6 Model complexity and FP32 inference measurements for the ablation variants. MACs were profiled using THOP package. Inference time and peak GPU memory were measured per CT volume.

Variant	Params (M)	MACs (G)	Time (s/volume)	Peak memory (GiB)
A0	42.5952	669.62	1.6952	1.9332
A1	29.2877	666.24	1.5192	1.8847
A2	41.1706	1501.56	2.7876	2.2198
A3	56.3657	1512.02	3.1499	2.2771
A4	56.3659	1507.33	2.9231	2.2764
A4b	51.3935	676.40	1.6729	1.9703
A5	56.6203	1517.82	2.9759	2.2763
A5_patch8	56.3909	1549.79	3.2909	2.2844

Adding the fixed-resolution Path-2 increased MACs from 666.24 G in A1 to 1501.56 G in A2, and inference time from 1.52 to 2.79 s per volume. Compared with standard CA (A3), GCA (A4) reduced measured inference time from 3.15 to 2.92 s per volume, while total MACs decreased only slightly. The module-level results are reported in Section 3.3.

A4b required 676.40 G MACs and 1.67 s per volume, compared with 1517.82 G and 2.98 s for A5. The higher cost of A5 mainly comes from upsampling and decoding Path-2 features at shallower stages, especially the high-resolution 3D convolutions. Reducing the Path-2 patch size to 8^3 further increased MACs to 1549.79 G and inference time to 3.29 s per volume, without improving DSC.

3.2 GSA and Boundary Accuracy

The A0–A1 comparison shows that higher tumor overlap does not always correspond to smaller surface distances. Over Dataset A, replacing SRA with GSA increased mean DSC from 0.6160 to 0.6443 and recall from 0.5331 to 0.5652. However, mean HD95 increased from 27.12 to 67.21 mm and ASSD from 7.34 to 11.07 mm. DSC and recall measure overlap and tumor coverage, whereas HD95 and ASSD measure the distances between predicted and annotated surfaces. A prediction can therefore recover more tumor voxels while retaining distant boundary errors.

One case illustrates this difference. From A0 to A1, DSC increased from 0.2607 to 0.5673 and recall from

0.1655 to 0.4549, but HD95 increased from 85.00 to 336.98 mm. With the full A5 configuration, DSC reached 0.7635 and HD95 decreased to 25.00 mm, with an ASSD of 5.43 mm. This example shows why overlap and surface distance should be assessed together. Across Dataset A, A5 achieved a mean DSC of 0.7424, HD95 of 21.60 mm, and ASSD of 4.59 mm, improving both overlap and boundary accuracy relative to the single-path GSA model.

3.3 GCA Computation

A2, A3, and A4 use the same dual-path design to compare no cross-attention, standard CA, and GCA. The A3–A4 comparison isolates the effect of replacing standard CA with GCA on attention computation. Table 7 reports both module-level estimates and whole-model measurements.

Table 7 Attention computation and inference efficiency for standard CA (A3) and GCA (A4). Core attention MACs include the QK^{T} and attention–value products. Attention-block estimates also include projections and compression. Whole-model MACs are obtained using THOP. Cumulative attention-map volume is an FP32 storage estimate, not measured peak GPU memory.

Metric	Standard CA (A3)	GCA (A4)	Change
Core attention MACs (G)	15.0324	3.2883	−78.13%
Estimated attention-block operations (G)	22.8542	6.3913	−72.03%
Cumulative attention-map volume (GiB, FP32)	0.4375	0.0957	−78.13%
Whole-model MACs (G)	1512.0233	1507.3335	−0.31%
Inference time (s/volume, FP32)	3.1499	2.9231	−7.20%
Peak GPU memory (GiB)	2.2771	2.2764	$<0.1\%$ reduction

GCA reduced core attention MACs from 15.03 to 3.29 G (78.13%). When projections, compression, and other attention-block operations were included, the estimated cost decreased from 22.85 to 6.39 G (72.03%). The estimated cumulative FP32 attention-map volume also decreased from 0.4375 to 0.0957 GiB.

For the complete network, THOP-reported MACs decreased from 1512.02 to 1507.33 G (0.31%). The smaller whole-model change is expected because the encoder and decoder account for much of the computation, and the separate attention-block estimates do not use exactly the same profiling scope as THOP. Measured FP32 inference time decreased from 3.15 to 2.92 s per volume (7.20%), while peak GPU memory was nearly unchanged. A3 and A4 also achieved similar segmentation results, with DSC values of 0.6559 and 0.6514, respectively. GCA therefore reduced the cost of cross-attention and shortened measured inference time while retaining comparable segmentation accuracy.

3.4 ERF Analysis

3.4.1 ERF Calculation

We calculated effective receptive fields (ERFs) to measure how strongly different regions of the input CT influenced features at selected encoder stages. CT images and GTV annotations from Dataset A were resampled to $1.87 \times 1.87 \times 5.0$ mm. A $128 \times 128 \times 64$ voxel patch was extracted around each annotated GTV. CT intensities were clipped to $[-175, 250]$ HU and normalized to $[0, 1]$. The GTV annotation was used to locate the feature positions and calculate distances for this analysis. It was not provided as an input to the network.

At each stage and path, feature values at positions corresponding to the GTV were averaged across spatial positions and channels to obtain a scalar probe s. The absolute gradient of s with respect to the input image I was used as the ERF response $E(\mathbf{x})$. The response within a region A was calculated as

$$E(\mathbf{x}) = \left| \frac{\partial s}{\partial I(\mathbf{x})} \right|, \tag{2}$$

$$S(A) = \sum_{\mathbf{x} \in A} E(\mathbf{x}). \tag{3}$$

No per-case min–max normalization was applied to the gradient responses.

The valid CT input field U excludes padded positions. A body mask $B \subseteq U$ was generated from the CT intensities in HU by thresholding at -500 HU, retaining the largest connected component, applying two iterations of three-dimensional morphological closing, and filling enclosed holes. This mask defines the approximate body region within the patch. Let T denote the annotated GTV and $d(\mathbf{x}, T)$ the shortest physical distance from a voxel to T, with distance zero inside the GTV. The cumulative neighborhood and response fractions are

$$N(r) = \{\mathbf{x} \in B : d(\mathbf{x}, T) \leq r\}, \tag{4}$$

$$F_{\text{body}}(r) = \frac{S(N(r))}{S(B)}, \tag{5}$$

$$F_{\text{full}}(r) = \frac{S(N(r))}{S(U)}, \tag{6}$$

$$P_{\text{out}} = \frac{S(U \setminus B)}{S(U)}. \tag{7}$$

Fig. 6 shows the regions used in these calculations. We use cumulative intrabody response for $F_{\text{body}}(r)$, the fraction of response inside the body mask that falls within r mm of the GTV. $F_{\text{full}}(r)$ uses the entire valid input field as its denominator, and P_{out} measures the response fraction outside the body mask. The 10-, 20-, and 30-mm distances are used for spatial analysis and are not clinical target margins. We calculated these measures for each case and used 2,000 paired patient-level bootstrap resamples to estimate 95% confidence intervals (CIs).

3.4.2 GSA and Body-Region Response

We first compared the Stage-1 Path-1 ERFs of A0 and A1. After replacing SRA with GSA, the fraction of response outside the body mask decreased by 10.28 percentage points (95% CI: -15.85 to -4.83), as shown in Fig. 7. When normalized by the entire input field, the cumulative response within 10–30 mm of the GTV increased by approximately 7.6–8.0 percentage points. However, the corresponding changes were small when normalized by the body-mask response, and their CIs included zero. These results suggest that GSA shifts the relative distribution of the ERF toward the body region, primarily by reducing the response outside the body rather than concentrating the intrabody response more tightly around the GTV. This shift indicates a greater emphasis on anatomically relevant image regions and is consistent with the improved DSC and recall observed in A1. Nevertheless, the increased HD95 and ASSD indicate that this redistribution does not necessarily lead to more accurate tumor boundaries.

3.4.3 Stage-1 Response by Encoder Path

Fig. 8 compares the Stage-1 cumulative intrabody responses of the two encoder paths in A4. Path-1 accumulated 73.06% of its response within the GTV and 84.11% within 20 mm of the GTV, whereas the corresponding fractions for Path-2 were 9.65% and 53.27%. These differences indicate that Path-1 exhibits a more tumor-centered response, while Path-2 distributes a greater proportion of its response over the surrounding anatomical regions. The distinct spatial response patterns suggest that the hierarchical Path-1 and fixed-resolution Path-2 provide complementary sensitivity to local tumor information and broader anatomical context, supporting their integration within the dual-path encoder.

3.4.4 A4b−−A5 ERF Comparison

We compared A4b and A5 to examine how their shallow encoder responses differed after training. In Stage-1 Path-1, A5 had higher cumulative intrabody response within 10, 20, and 30 mm of the GTV by 10.92, 10.72, and 10.44 percentage points, respectively (Fig. 9). The paired 95% CIs were [6.67, 15.21], [6.50, 14.95], and [6.35, 14.57] percentage points. Most of the increase came from the response within the GTV itself, which rose from 65.45% in A4b to 75.60% in A5.

Path-2 changed differently. Its cumulative response within 30 mm decreased by 3.50 percentage points (95% CI: -5.44 to -1.55), while the changes at 10 and 20 mm were smaller. In A5, the shallow Path-1 response

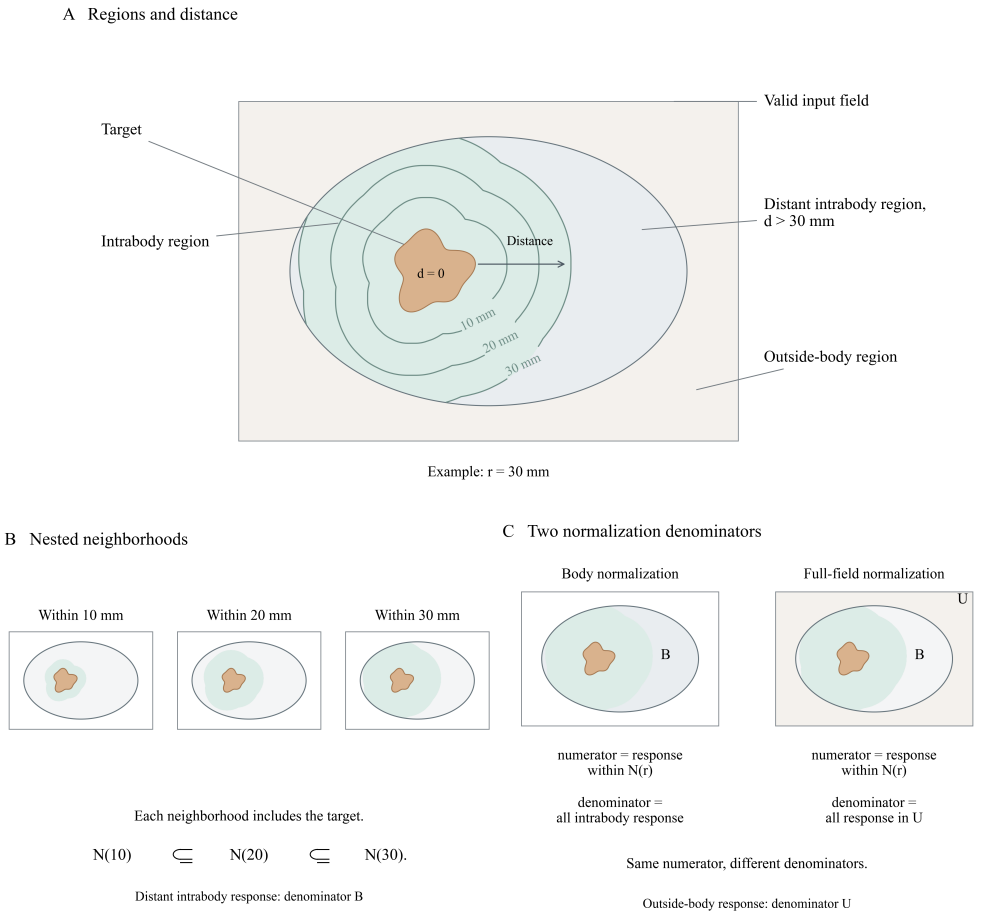

Figure 6 Regions used to calculate the ERF. U is the valid CT input field, B is the automatically generated body mask, and T is the annotated GTV. The 10-, 20-, and 30-mm neighborhoods include the GTV and extend outward in three dimensions. Response fractions use either the body-mask response or the full input-field response for normalization.

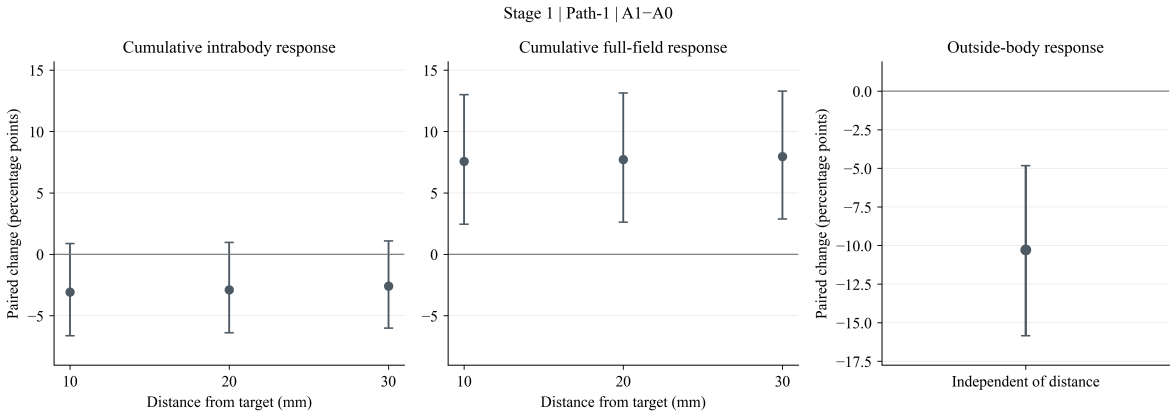

Figure 7 Changes in Stage-1 Path-1 ERF after replacing SRA with GSA (A1 minus A0). Left: cumulative intrabody response. Middle: cumulative response normalized by the full CT input field. Right: response fraction outside the body mask. Points show mean paired differences. Error bars indicate 95% bootstrap confidence intervals. Fraction changes are reported in percentage points.

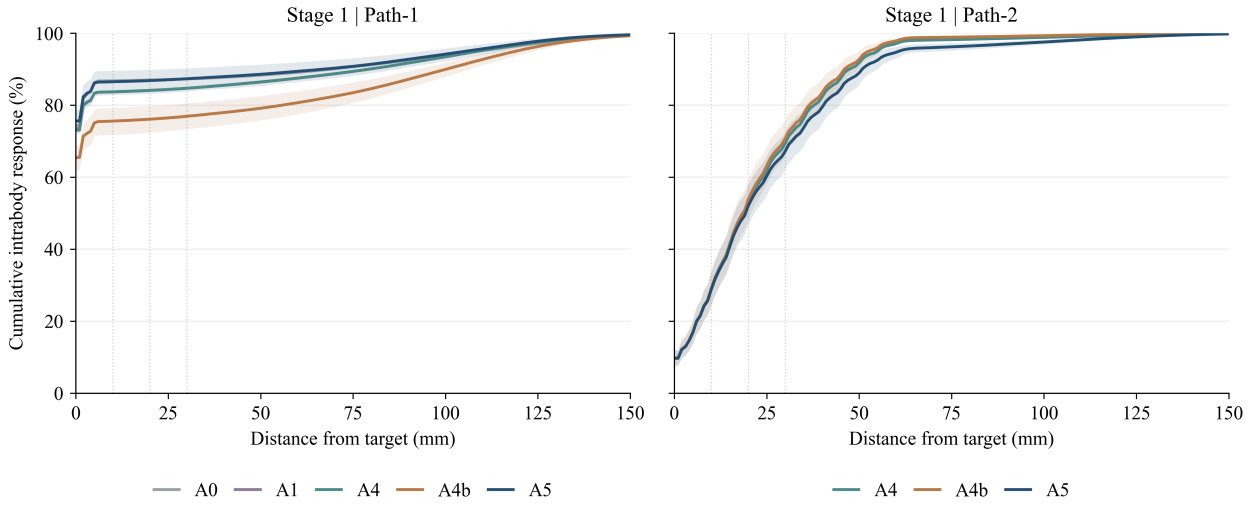

Figure 8 Stage-1 cumulative intrabody response plotted against the physical distance from the GTV for both encoder paths. Each curve is normalized by the total ERF response inside the body mask. Shaded bands indicate pointwise 95% confidence intervals for the highlighted configurations.

was therefore more concentrated near the GTV, while Path-2 retained a greater share of its response at larger distances.

3.4.5 Stage-4 Response

Fig. 10 shows that the Stage-4 ERFs extended over larger distances from the GTV than the shallow Path-1 responses. In A5, Path-1 and Path-2 accumulated only 9.16% and 5.35% of their intrabody responses within 30 mm of the GTV, respectively. At 100 mm, these fractions increased to 61.94% and 60.07%, indicating that substantial responses were distributed beyond the immediate tumor neighborhood. Unlike the more localized responses observed in shallow Path-1 features, both paths exhibited broader spatial sensitivity at Stage 4. This stage-dependent pattern suggests that the encoder can represent tumor-proximal information at shallow levels while incorporating a wider anatomical context at deeper levels, providing complementary spatial information across the encoding stages.

4 Temporal Testing and Failure Modes

4.1 Dataset B Performance

Fig. 11 shows the patient-level DSC distributions of the 19 methods on Dataset B. The full quantitative results are reported in the main manuscript. CFINet achieved a mean DSC of 0.7116 and a median DSC of 0.7587, with a minimum of 0.5088. Its distribution was concentrated toward higher DSC values, whereas several baselines showed substantially lower scores in individual cases. nnUNet (cascade) also had no DSC below 0.50, although its mean DSC was lower at 0.6839. PHTrans and UNETR++ showed wider differences between their best- and worst-performing cases. The boxplot complements the mean scores in the main manuscript by showing the variation across individual cases.

4.2 Cohort Characteristics

Datasets A and B were collected at the same institution using the same CT scanner and acquisition protocol, but at different times. Both datasets have a 512×512 image matrix, an in-plane pixel spacing of approximately 0.98 mm, and a slice thickness of 5 mm. Air-noise measurements were similar between the cohorts, with median standard deviations of 33.36 and 33.30 HU in Datasets A and B, respectively. The median intensity within the annotated GTV was also similar (39.0 versus 39.5 HU). None of the 79 measured CT image features

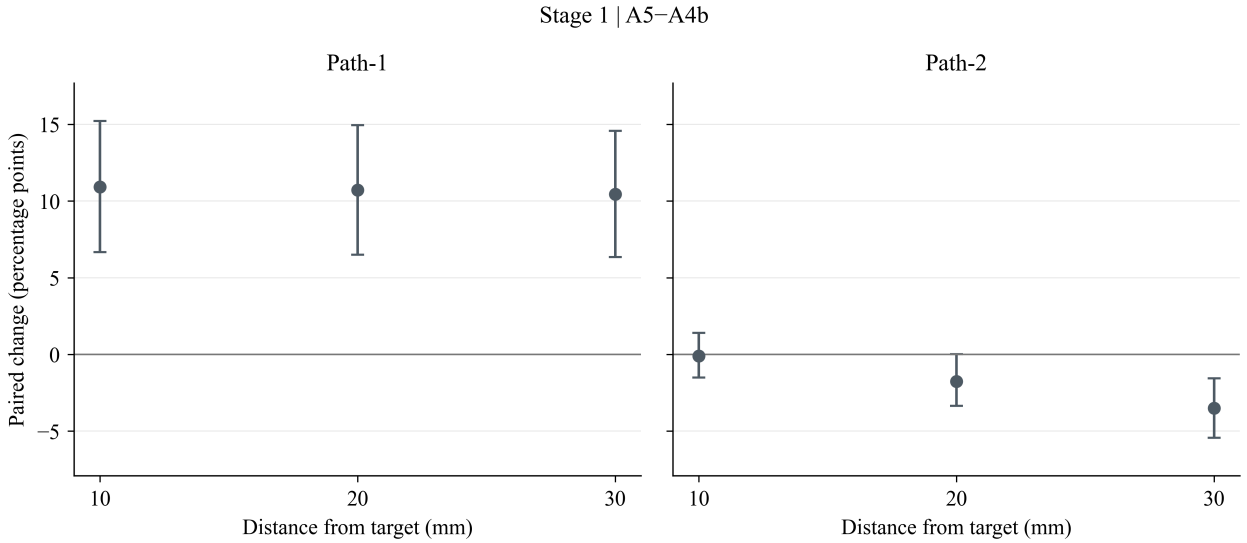

Figure 9 Differences in Stage-1 cumulative intrabody response between A5 and A4b, calculated as A5 minus A4b. Each neighborhood includes the GTV and surrounding body-mask voxels within the specified distance. Points show mean paired differences, with 95% bootstrap confidence intervals.

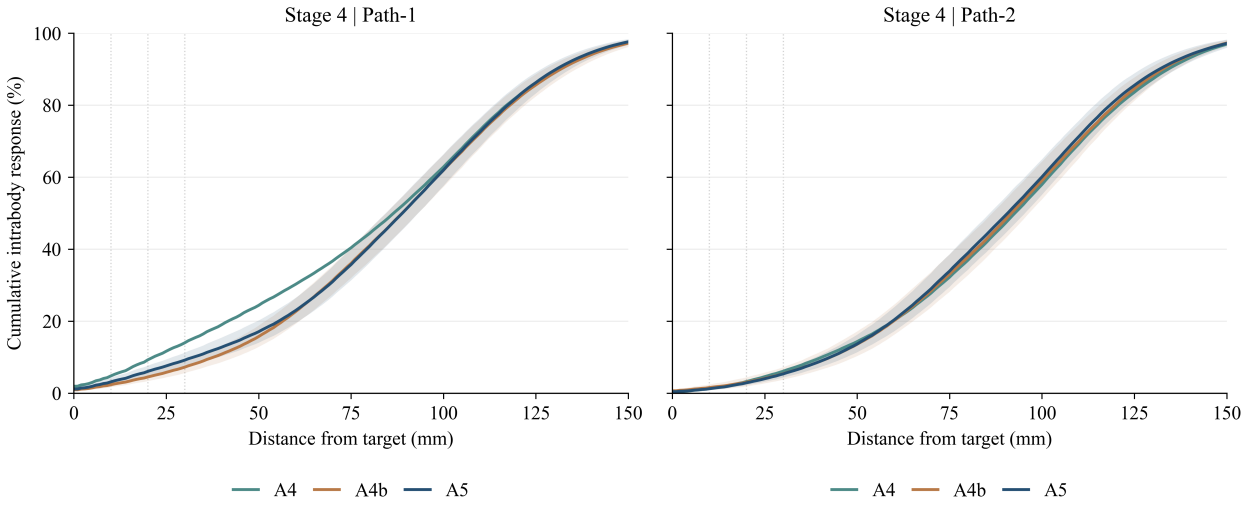

Figure 10 Stage-4 cumulative intrabody response plotted against the physical distance from the GTV for both encoder paths. Compared with Stage-1 Path-1, the deep features show a greater share of their response at larger distances. Shaded bands indicate pointwise 95% confidence intervals for the highlighted configurations.

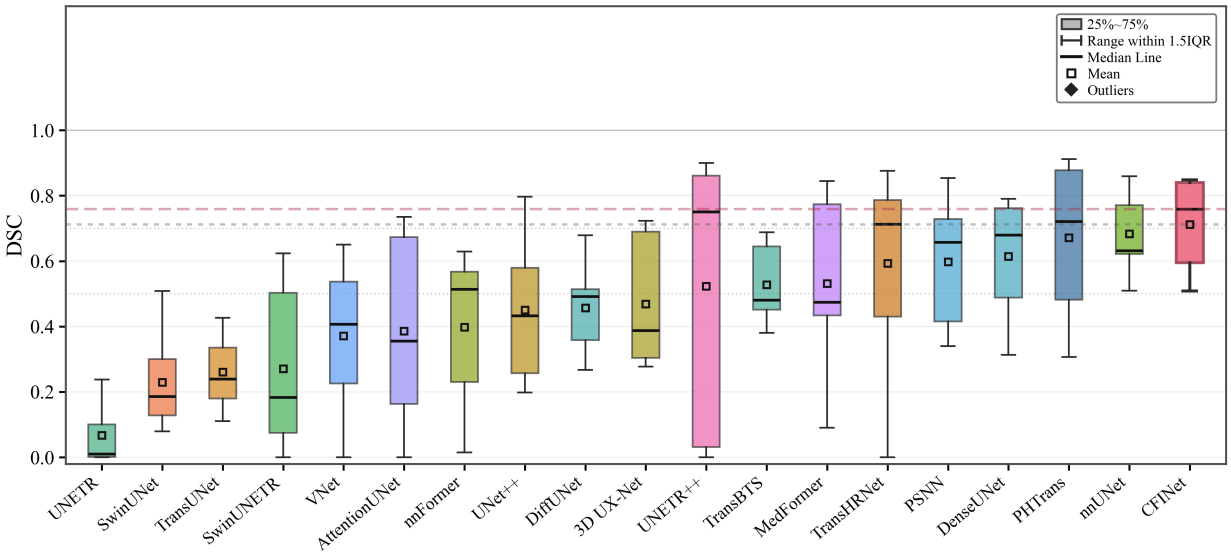

Figure 11 Patient-level DSC distributions of the 19 segmentation methods on the temporally held-out Dataset B. Boxes show the interquartile range, horizontal lines indicate medians, and open squares indicate means. Whiskers extend to observations within 1.5 times the interquartile range, and filled diamonds indicate outliers.

showed a statistically significant between-cohort difference after correction for multiple comparisons. The measured image features did not show significant differences after multiple-comparison correction. However, the small temporal cohort may limi the ability of these tests to detect between-cohort differences.

We also compared the size and shape of the reference GTVs (Table 8). Dataset B had a lower median mean axial GTV area than Dataset A (469.5 versus 598.9 mm^2) and a smaller median axial minor-axis length (19.20 versus 22.46 mm). In contrast, the median superior–inferior extent was similar (67.5 versus 70.0 mm), and the median axial aspect ratio was 1.491 in both cohorts. Fig. 12 shows the patient-level distributions of axial area and minor-axis length. Although the median transverse dimensions were smaller in Dataset B, neither difference was statistically significant.

Table 8 Selected CT characteristics and reference GTV geometry in Datasets A and B. Values are cohort medians. The reported p values are exploratory between-cohort comparisons of the selected measures.

Measure	Dataset A	Dataset B	p value
In-plane pixel spacing (mm)	0.9766	0.9766	0.688
Slice thickness (mm)	5.0	5.0	0.436
Air-noise standard deviation (HU)	33.36	33.30	0.876
Median GTV intensity (HU)	39.0	39.5	0.907
GTV volume (mL)	43.37	34.95	0.353
Superior–inferior GTV extent (mm)	70.0	67.5	0.930
Mean axial GTV area (mm^2)	598.9	469.5	0.270
Mean axial minor-axis length (mm)	22.46	19.20	0.296
Axial cross-sectional aspect ratio	1.491	1.491	0.680

4.3 Dataset A Failure Modes

We examined patient-level predictions on Dataset A to identify errors that were not fully reflected by mean DSC. The inspected predictions showed three main error patterns: incomplete tumor coverage, extension beyond the annotated GTV, and displaced or disconnected predictions. Empty predictions were also found in some baseline methods. These errors can lead to different combinations of overlap and surface-distance measurements.

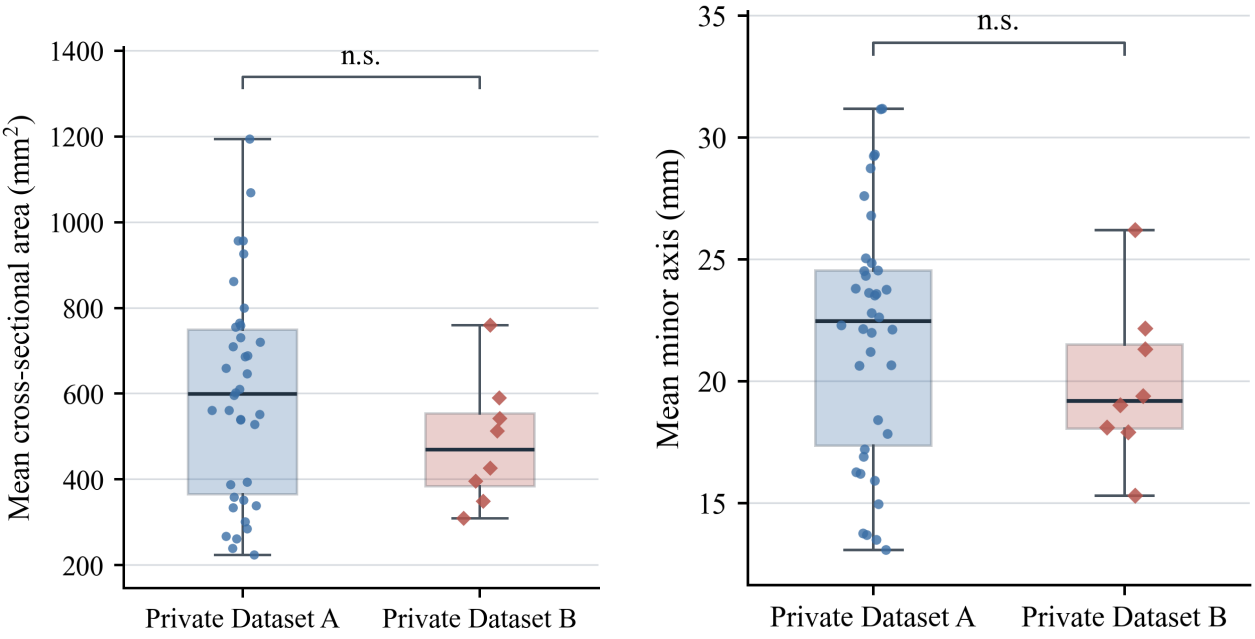

Figure 12 Comparison of reference GTV transverse dimensions between Datasets A and B. Left: mean axial GTV area. Right: mean axial minor-axis length. Each point represents one case. The observed between-cohort differences were not statistically significant.

Incomplete coverage was particularly evident in methods with high precision but relatively low recall. For example, UNETR++ achieved the highest mean precision (0.8577), but its recall was 0.5671. In a representative small-GTV case, UNETR++ achieved a DSC of 0.0268 and a recall of 0.014, compared with a DSC of 0.7023 and a recall of 0.660 for CFINet. Complete omission was observed in another case, shown in Fig. 13. TransHRNet and UNETR++ produced empty predictions, whereas CFINet and nnUNet (cascade) identified the target, with DSC values of 0.6948 and 0.7376. Although nnUNet (cascade) achieved higher overlap in this example, CFINet had a lower HD95 (6.12 versus 11.47 mm). These cases illustrate different degrees of missed tumor coverage, ranging from a small prediction to no detected target.

Large tumors presented another difficulty. Fig. 14 shows a case with a reference GTV volume of 49.3 mL in which all seven selected methods had limited recall. CFINet achieved a DSC of 0.4772 and a recall of 0.320, indicating that substantial portions of the GTV remained undetected. MedFormer had a similar DSC of 0.4895, but the two predictions differed in spatial distribution. CFINet achieved higher precision (0.940 versus 0.527), a smaller centroid displacement (16.0 versus 69.8 mm), and lower HD95 (23.72 versus 48.22 mm). The CFINet prediction contained three connected components, compared with 16 for MedFormer. The comparison illustrates why DSC alone cannot describe the spatial distribution of errors.

Boundary errors were also observed when the predicted GTV extended beyond the annotated region. In the case shown in Fig. 15, the selected methods generally had high recall but lower precision, consistent with over-segmentation. CFINet and nnUNet (cascade) achieved similar DSC values (0.6926 and 0.6818) and similar predicted-to-reference volume ratios (1.523 and 1.524). However, their HD95 values differed markedly (17.95 versus 93.63 mm). CFINet produced one connected component, compared with four for nnUNet (cascade). Other methods achieved higher DSC values in this particular case, but the comparison between CFINet and nnUNet (cascade) illustrates how similar overlap and volume can coexist with different boundary errors.

Across Dataset A, CFINet achieved the highest mean DSC (0.7424) and recall (0.7267), together with the lowest mean HD95 (21.60 mm) and ASSD (4.59 mm) among the 19 methods. For comparison, DenseUNet achieved a DSC of 0.6773 but had a mean HD95 of 115.92 mm, while SwinUNETR achieved a recall of 0.7075 with a mean HD95 of 209.15 mm. These differences show that obtaining substantial tumor overlap does not

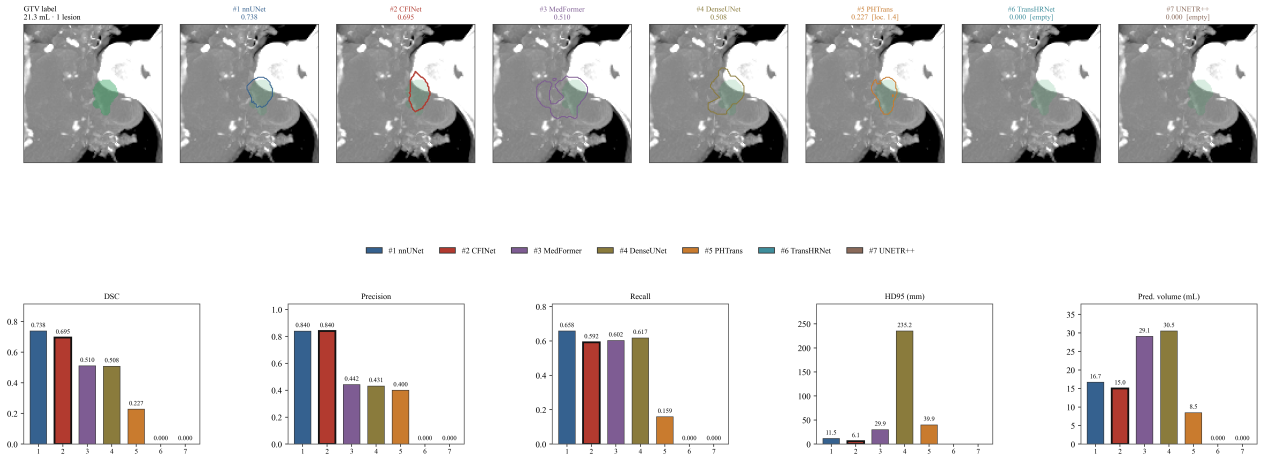

Figure 13 Representative Dataset A case illustrating incomplete GTV detection. Predicted contours are shown with the reference GTV on a CT slice. TransHRNet and UNETR++ produced empty predictions, while CFINet and nnUNet (cascade) identified the target. Patient-level DSC, precision, recall, HD95, and predicted volume are shown below the images.

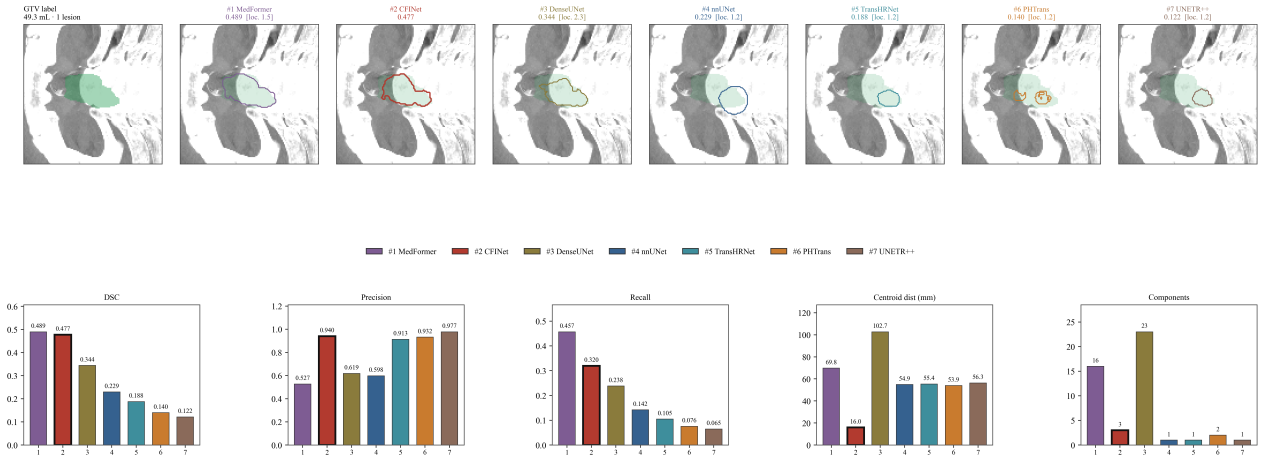

Figure 14 Representative large-GTV case from Dataset A. The upper row compares predicted contours with the reference GTV. The lower panels report DSC, precision, recall, centroid displacement, and the number of connected components. CFINet and MedFormer achieved similar DSC values but differed in the location and fragmentation of their predictions.

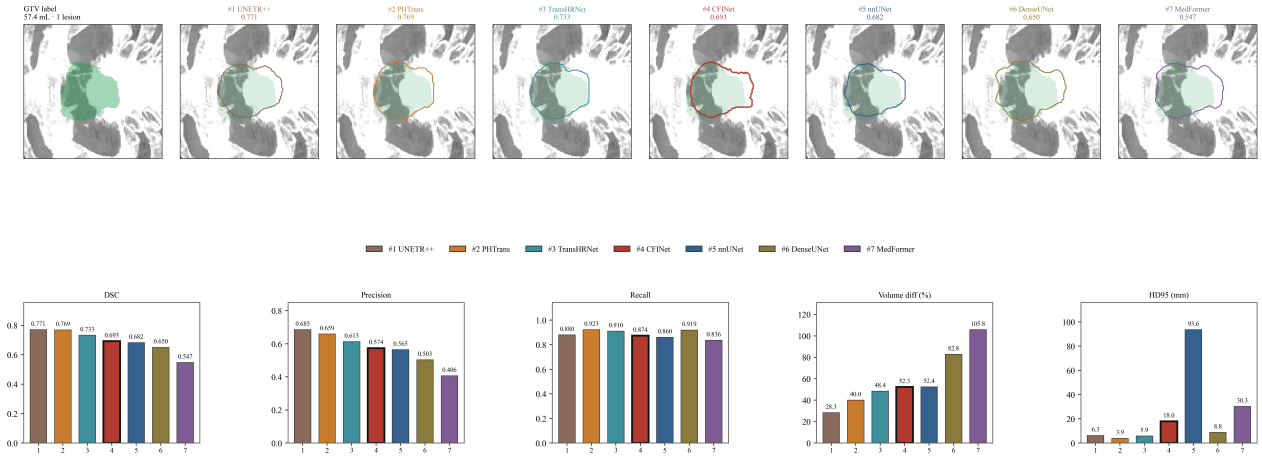

Figure 15 Representative Dataset A case with predictions extending beyond the reference GTV. Predicted contours and patient-level measurements are shown for seven methods. The lower panels compare DSC, precision, recall, relative predicted volume, and HD95. CFINet and nnUNet (cascade) had similar DSC and predicted volumes but different HD95 values.

always prevent large boundary errors. CFINet still missed tumor regions in some patients, as the large-GTV example demonstrates, but its overall results indicate a more favorable balance between tumor coverage and surface accuracy.

4.4 Dataset B Failure Modes

The patient-level DSC heatmap in Fig. 16 shows how the models performed on the subsequently collected Dataset B cases. Several methods achieved good overlap on some patients but performed poorly on others. CFINet and nnUNet (cascade) were the only methods with DSC above 0.50 in every tested case. Their minimum DSC values were 0.5088 and 0.5097, respectively. By comparison, the minimum DSC was 0.3064 for PHTrans and 0.3128 for DenseUNet, and some other methods produced predictions with almost no overlap.

Fig. 17 compares each model's mean DSC with its lowest patient-level DSC. CFINet had the highest mean DSC of 0.7116 and a minimum of 0.5088. Its largest patient-level HD95 was 35.0 mm, compared with 161.4 mm for nnUNet (cascade) and 51.69 mm for PHTrans. CFINet also achieved the highest mean DSC (0.6324) among the five cases where the average DSC across the 19 models was below 0.50, followed by nnUNet (cascade) (0.6036) and PHTrans (0.5405). This subgroup comparison is descriptive and was based on the observed results, rather than a predefined difficulty classification.

Predicted tumor volume provided further information about the low-overlap cases. DenseUNet and PHTrans had median predicted-to-reference GTV volume ratios of 0.587 and 0.680, respectively, with recall values of 0.4955 and 0.5939. Their predictions were therefore often smaller than the annotated targets. In contrast, DiffUNet had a median volume ratio of 1.941 and a precision of 0.3625, indicating substantial false-positive prediction. CFINet and nnUNet (cascade) both had median volume ratios of approximately 0.855. These measurements describe different tendencies toward under-segmentation and over-segmentation, although predicted volume alone does not determine boundary accuracy.

Individual difficult cases also showed that CFINet was not uniformly the best method. In one case with an elongated axial GTV, CFINet achieved a DSC of 0.5088, below nnUNet (cascade) and PHTrans. Its precision was 0.927 but recall was 0.351, indicating incomplete tumor coverage. In another difficult case, 17 of the 19 methods had DSC below 0.50, while CFINet and nnUNet (cascade) achieved 0.5136 and 0.5097, respectively. These examples show both the remaining difficulty of delineating some tumors and the differences in how methods respond to challenging cases.

Compared with Dataset A, CFINet's mean DSC decreased from 0.7424 to 0.7116 on Dataset B. This is a descriptive comparison between different patients and different trained model instances. The cohort analysis

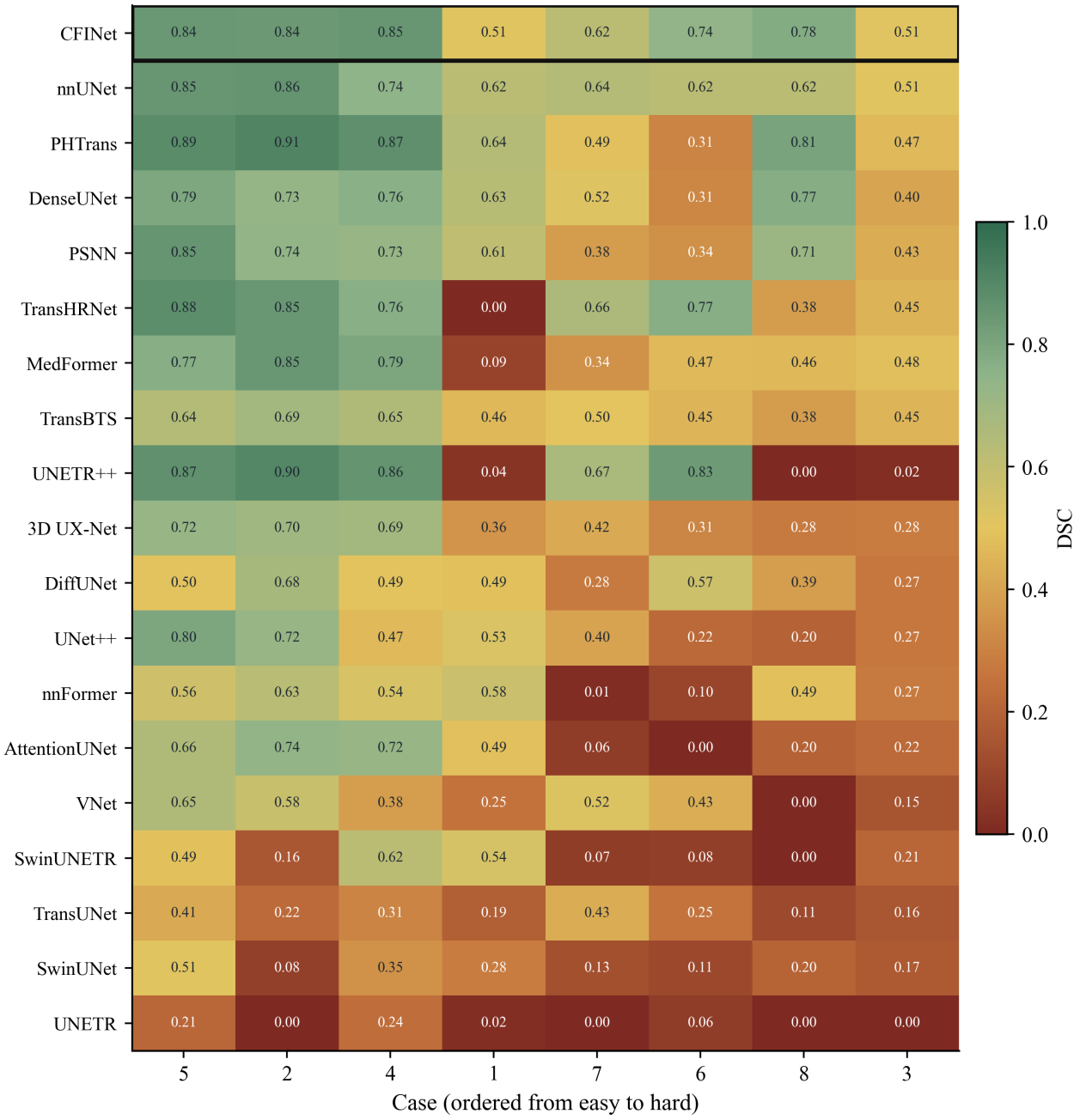

Figure 16 Patient-level DSC of all 19 segmentation methods on Dataset B. Columns represent cases ordered by the mean DSC across models, and rows represent models. Green indicates higher overlap and red indicates lower overlap.

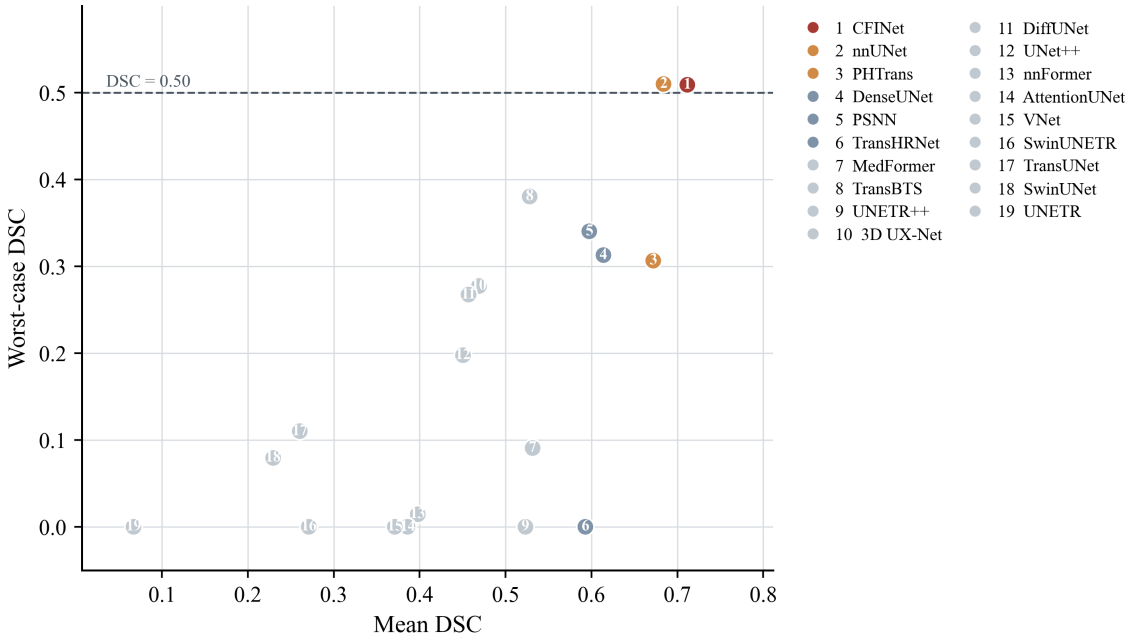

Figure 17 Mean DSC and lowest patient-level DSC of each method on Dataset B. The dashed horizontal line indicates DSC = 0.50. Higher values on both axes indicate greater overlap on average and in the lowest-scoring case.

found no statistically significant differences among the measured CT image features after multiple-comparison correction, and the observed transverse GTV size differences were also not statistically significant. The available measurements do not identify a specific cause of the difference. Nevertheless, CFINet retained the highest mean DSC and favorable surface-distance metrics relative to the compared methods on Dataset B.

5 GTV-to-CTV/PTV Geometry

5.1 Expansion and Evaluation

We examined whether differences in GTV segmentation were reflected in the geometry of expanded target regions. The selected margins were informed by esophageal radiotherapy contouring recommendations, which describe longitudinal GTV-to-CTV extensions of approximately 2–4 cm and radial extensions of approximately 0.5–1 cm, depending on the clinical setting [4, 17, 19]. The PTV-related settings included additional geometric margins to reflect the need to accommodate setup uncertainty and motion [4, 9]. We used these fixed margins for geometric comparisons, not for patient-specific clinical target delineation.

Let G denote the physician-defined GTV, $\hat{G}_m$ the predicted GTV from model m, and C and P the physician-defined CTV and PTV. For a longitudinal margin z and an in-plane margin r, the expansion was defined as

$$\mathcal{E}_{z,r}(M) = \mathrm{Dilate}_z(\mathrm{Dilate}_{xy,r}(M)) \,. \tag{8}$$

In-plane dilation used physical Euclidean distances within each axial CT slice. Longitudinal dilation was applied along the superior–inferior slice axis, with the requested distance converted to the nearest integer number of slices. All predictions and references were processed in the same image geometry.

Clinical CTV and PTV contours cannot generally be reproduced by fixed GTV margins alone. We therefore evaluated the expanded predictions against the portions of the clinical targets within the corresponding expanded reference GTV:

$$C_{z,r} = C \cap \mathcal{E}_{z,r}(G), \tag{9}$$

$$P_{z,r} = P \cap \mathcal{E}_{z,r}(G). \tag{10}$$

For $T_{z,r} \in \{C_{z,r}, P_{z,r}\}$ and $\hat{T}_{m,z,r} = \mathcal{E}_{z,r}(\hat{G}_m)$, we calculated

$$\mathrm{DSC}_{m,z,r} = \frac{2|\hat{T}_{m,z,r} \cap T_{z,r}|}{|\hat{T}_{m,z,r}| + |T_{z,r}|}, \tag{11}$$

$$\mathrm{Coverage}_{m,z,r} = \frac{|\hat{T}_{m,z,r} \cap T_{z,r}|}{|T_{z,r}|}. \tag{12}$$

To characterize expansion beyond the physician-defined PTV, we also measured

$$V_{\mathrm{out},m,z,r} = \mathrm{Vol}\left(\hat{T}_{m,z,r} \setminus P\right). \tag{13}$$

Thus, Dice quantifies agreement with the local clinical target, coverage measures the proportion of that target included in the expanded prediction, and V_{out} measures expanded volume outside the clinical PTV. The last measure is a geometric volume, not delivered radiation dose.

5.2 Reference GTV Expansion

We additionally expanded the physician-defined GTV using the same margins and computed its Dice against the local clinical target:

$$D_{\mathrm{GT\text{-}exp},z,r} = \mathrm{DSC}(\mathcal{E}_{z,r}(G), T_{z,r}). \tag{14}$$

This reference measures how closely a fixed expansion of the annotated GTV matches the local clinical target. It is not an upper bound on expanded-prediction Dice. An exact GTV prediction need not yield a Dice of one because the clinical target may differ from the fixed expansion. Conversely, a more conservative prediction can sometimes obtain a higher local-target Dice than the reference expansion.

Fig. 18 compares the mean Dice of reference-GTV expansion and CFINet predictions under identical margin settings. Within each cohort and setting, both means were calculated from the same set of evaluable cases. For the 30×10 mm CTV setting, the reference-expansion Dice was 0.7819 on Dataset A and 0.7175 on Dataset B, compared with 0.6743 and 0.7074 for CFINet, respectively. Under the 35×15 mm PTV setting, the corresponding reference-expansion Dice values were 0.8334 and 0.7854, compared with 0.7160 and 0.7636 for CFINet. These results indicate that the agreement between fixed GTV expansion and clinical target geometry varies across the two cohorts. The reference-expansion Dice therefore serves as a geometric benchmark for interpreting model performance rather than an upper bound on the Dice of expanded predictions.

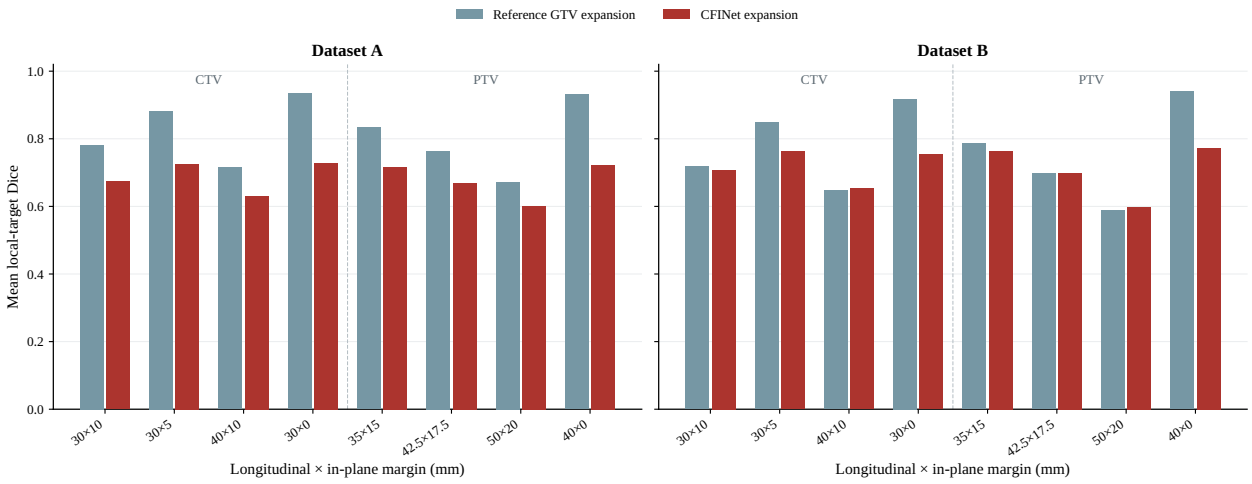

Figure 18 Comparison of mean local-target Dice for fixed expansions of the reference GTV and CFINet predictions on Datasets A and B. The eight settings are expressed as longitudinal × in-plane margins in millimeters. Reference-GTV expansion provides a geometric comparison, not a theoretical upper bound on model Dice.

5.3 GTV-to-CTV Expansion

We first fitted longitudinal and in-plane margins to the physician-defined CTV contours to estimate the typical expansion distances from the reference GTV. This analysis was intended to characterize the clinical GTV-to-CTV relationship and assess its consistency with published contouring recommendations [4, 17, 19]. The median fitted longitudinal margin was 25 mm (interquartile range [20, 30] mm), and the median in-plane margin was 7.5 mm (interquartile range [7.5, 10.0] mm), with a median fitted Dice of 0.680. These distances were broadly consistent with the recommended expansion ranges. The fitting was used only for this descriptive analysis, while subsequent model comparisons employed the same predefined expansion settings for all methods.

For model comparisons, four fixed settings were evaluated: 30×10, 30×5, 40×10, and 30×0 mm, with longitudinal margin listed first. The first three settings examine longitudinal and radial ranges informed by the contouring literature [4, 17, 19]. The 30×0 mm setting isolates the effect of longitudinal dilation. All models were evaluated with the same settings.

On Dataset A, CFINet achieved the highest mean Dice across all four CTV expansion settings when evaluated against the GTV-centered portions of the physician-defined CTV contours (Table 9). At 30×10 mm, its Dice was 0.6743, compared with 0.6708 for PHTrans and 0.6285 for nnUNet (cascade). Reducing the in-plane margin to 5 mm increased the CFINet Dice to 0.7250. These results reflect geometric agreement with the local clinical CTV rather than the accuracy of complete CTV delineation. The differences between CFINet and PHTrans were small.

5.4 GTV-to-PTV Expansion

Table 9 Mean Dice of expanded GTV predictions against the local clinical CTV and PTV reference regions on Dataset A. Margins are longitudinal × in-plane distances in millimeters. Bold denotes the highest mean Dice in each setting.

	Local CTV				Local PTV			
Model	30 × 10	30 × 5	40 × 10	30 × 0	35 × 15	42.5 × 17.5	50 × 20	40 × 0
CFINet	**0.6743**	**0.7250**	**0.6308**	**0.7288**	**0.7160**	**0.6690**	0.6011	**0.7224**
nnUNet (cascade)	0.6285	0.6740	0.5876	0.6862	0.6616	0.6201	0.5580	0.6771
PHTrans	0.6708	0.7165	0.6275	0.7204	0.7100	0.6669	0.6016	0.7121
TransHRNet	0.6502	0.6850	0.6130	0.6850	0.6879	0.6516	0.5912	0.6820
UNETR++	0.6640	0.6806	0.6302	0.6592	0.6957	0.6639	**0.6077**	0.6511
DenseUNet	0.5285	0.5993	0.4861	0.6470	0.5569	0.5104	0.4520	0.6417
MedFormer	0.4414	0.5054	0.4058	0.5460	0.4678	0.4264	0.3746	0.5452

We further evaluated the expanded GTV predictions against the GTV-centered portions of the physician-defined PTV contours using four margin settings: 35×15, 42.5×17.5, 50×20, and 40×0 mm. Relative to the 30×10 and 40×10 mm CTV settings, the 35×15 and 50×20 mm settings increase both longitudinal and in-plane margins by 5 and 10 mm, respectively. The 42.5×17.5 mm setting provides an intermediate comparison, while 40×0 mm examines longitudinal expansion alone. All margins were applied directly to the predicted GTV masks. These settings were used to assess geometric sensitivity rather than to reproduce patient-specific clinical PTV prescriptions.

On Dataset A, CFINet achieved the highest mean Dice under the 35×15, 42.5×17.5, and 40×0 mm settings, with values of 0.7160, 0.6690, and 0.7224, respectively (Table 9). Under the largest 50×20 mm setting, UNETR++ achieved a slightly higher Dice (0.6077) than PHTrans (0.6016) and CFINet (0.6011). Fig. 19 compares the mean Dice across the CTV and PTV settings on both datasets. These results show that model rankings can vary with the expansion margins, although differences among the leading methods were small under several settings.

Fig. 20 summarizes the model rankings across the eight expansion settings on Dataset A. Among the seven evaluated methods, CFINet ranked first in seven settings, while UNETR++ achieved the highest mean Dice under the 50×20 mm PTV setting. This result further illustrates the generally consistent performance of CFINet across different expansion margins, while showing that its advantage was not uniform across all settings.

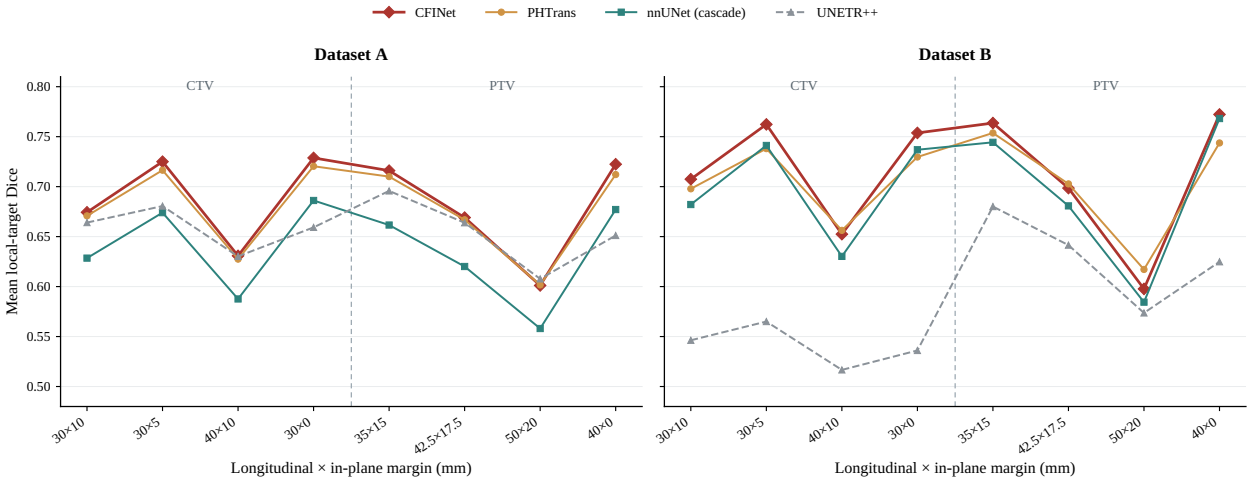

Figure 19 Mean local-target Dice across the four CTV and four PTV expansion settings on Dataset A and the temporally held-out, same-institution Dataset B. The two cohorts use the same expansion rules. Each setting is expressed as longitudinal × in-plane margin in millimeters. The plotted models are selected for clarity, while the corresponding tables provide additional comparisons.

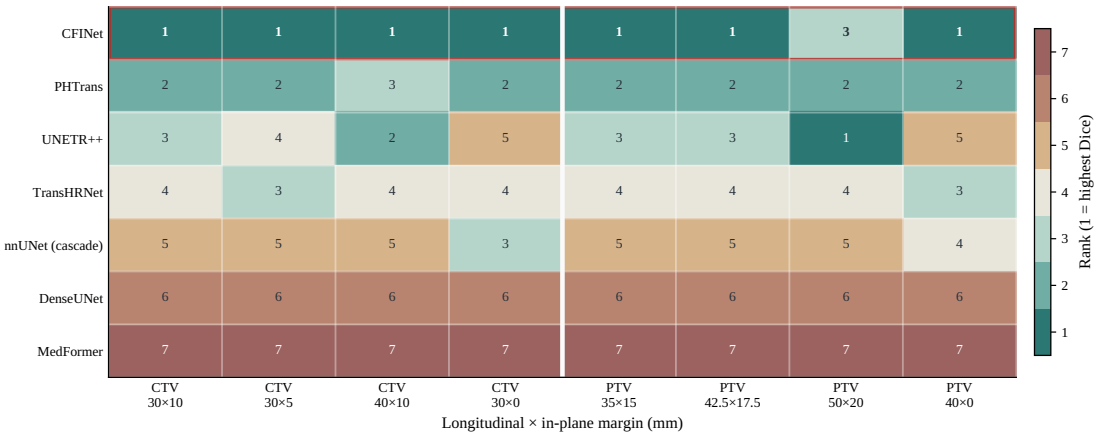

Figure 20 Ranking of the seven models included in the Dataset A geometric analysis, based on mean local-target Dice for each expansion setting. A rank of 1 indicates the highest mean Dice among these seven models. Margins are longitudinal × in-plane distances in millimeters. The heatmap does not represent rankings on Dataset B.

5.5 Coverage and Outside-PTV Volume

Dice alone cannot distinguish incomplete local-target coverage from excessive expansion beyond the clinical PTV. We therefore evaluated local-target coverage alongside outside-PTV volume under representative settings on Dataset A (Fig. 21). At the 30×10 mm CTV setting, CFINet achieved 86.8% coverage with an outside-PTV volume of 71.63 mL. Compared with nnUNet (cascade), CFINet provided higher coverage (86.8% vs. 84.9%) while reducing outside-PTV volume from 107.20 to 71.63 mL. PHTrans yielded lower coverage (84.2%) and less outside-PTV volume (58.19 mL). In contrast, DenseUNet and MedFormer achieved higher coverage (92.3% and 92.4%), but their outside-PTV volumes were substantially larger (219.06 and 358.43 mL).

At the 42.5×17.5 mm PTV setting, CFINet maintained a similar advantage over nnUNet, achieving higher coverage (90.6% vs. 88.0%) with less outside-PTV volume (284.18 vs. 350.53 mL). These results show that CFINet achieved a more favorable combination of local-target coverage and outside-PTV volume than nnUNet under both representative settings. The comparisons with other methods further highlight the trade-off between covering the local clinical target and limiting expansion beyond the clinical PTV. Although these findings support the geometric performance of CFINet, they do not establish clinical target acceptability or dosimetric benefits.

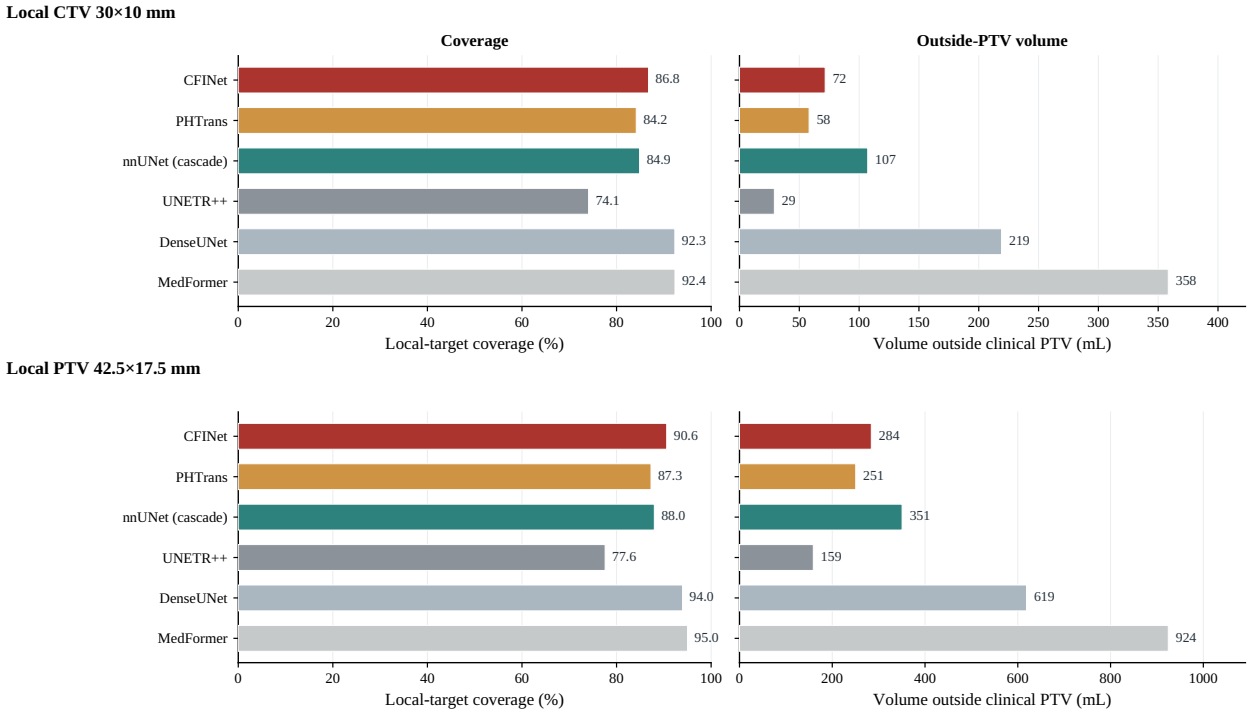

Figure 21 Comparison of local-target coverage and outside-PTV volume on Dataset A. The upper panels show the 30×10 mm CTV setting, and the lower panels show the 42.5×17.5 mm PTV setting. Left panels report local-target coverage (%), while right panels show expanded-mask volume outside the physician-defined PTV (mL). Each bar represents the model mean over its available evaluations. Higher coverage and lower outside-PTV volume represent distinct geometric objectives and do not directly indicate treatment-planning acceptability or delivered dose.

5.6 Dataset B Geometry

The same fixed-margin analysis was applied to the temporally held-out Dataset B, which was excluded from all stages of model development. Table 10 reports geometric Dice and local-target coverage under the representative 30×10 mm CTV and 35×15 mm PTV settings. CFINet achieved the highest mean Dice among the nineteen evaluated models in both settings. For local CTV, CFINet obtained a Dice of 0.7074 and coverage of 92.3%, compared with 0.6978 and 84.9% for PHTrans and 0.6821 and 87.7% for nnUNet (cascade). Under the PTV setting, CFINet achieved a Dice of 0.7636 and coverage of 93.3%, exceeding PHTrans (0.7537

and 85.5%) and nnUNet (0.7444 and 89.9%). Although the Dice differences among the leading methods were relatively small, CFINet consistently achieved higher local-target coverage than these two competitive baselines under both settings.

The margin-sensitivity results in Fig. 19 further demonstrate the performance of CFINet across different expansion settings on Dataset B. Among the nineteen models, CFINet achieved the highest mean Dice in five settings and ranked second in the remaining three. The differences among leading methods were small under several wider-margin settings, indicating that their relative performance depended partly on the expansion geometry. Nevertheless, CFINet maintained consistently competitive geometric agreement across the evaluated settings. Together with the representative CTV and PTV results, these findings suggest that the favorable geometric performance of CFINet was retained in the temporally held-out cohort, although the evaluation was conducted at the same institution rather than at an independent clinical site.

Table 10 Mean local-target Dice and coverage on Dataset B under representative CTV (30 × 10 mm) and PTV (35 × 15 mm) expansion settings. All values use the same longitudinal × in-plane margin definition. Bold denotes the highest mean Dice.

	Local CTV 30 × 10		Local PTV 35 × 15	
Model	Dice	Coverage (%)	Dice	Coverage (%)
CFINet	**0.7074**	92.3	**0.7636**	93.3
PHTrans	0.6978	84.9	0.7537	85.5
DenseUNet	0.6879	83.0	0.7293	84.1
nnUNet (cascade)	0.6821	87.7	0.7444	89.9
TransBTS	0.6354	81.8	0.7027	86.5
TransHRNet	0.6169	81.1	0.6643	84.2
VNet	0.5916	65.5	0.6855	74.7
UNETR++	0.5463	69.8	0.6801	81.8
AttentionUNet	0.5288	63.7	0.5738	67.3
3D UX-Net	0.4863	85.2	0.5318	90.2
UNet++	0.4721	82.0	0.4851	87.4
nnFormer	0.4252	76.3	0.4775	81.2
MedFormer	0.3965	90.5	0.4213	93.6
TransUNet	0.3893	77.8	0.4595	84.1
PSNN	0.3704	93.9	0.3861	96.4
DiffUNet	0.3182	91.5	0.3619	93.9
SwinUNet	0.3005	78.2	0.3660	82.2
SwinUNETR	0.2242	66.5	0.2737	80.2
UNETR	0.1189	41.0	0.1530	54.7

These evaluations were restricted to the GTV-centered portions of the physician-defined CTV and PTV contours. The results therefore reflect how GTV segmentation differences affect subsequent fixed-margin expansions rather than the accuracy of complete clinical target delineation. They provide additional support for investigating CFINet predictions as initial contours for physician review and refinement.

6 SegTHOR Evaluation

We evaluated CFINet on SegTHOR, a public benchmark for thoracic organs at risk (OARs), against 11 segmentation methods: TransUNet [2], SwinUNet [1], UNet++ [24], TransBTS [18], 3D UX-Net [11], nnFormer [23], nnUNet [7], UNETR [6], SwinUNETR [5], MedFormer [3], and DiffUNet [20]. The organ-wise comparisons are summarized in Tables 11 and 12.

CFINet achieved the highest mean DSC (0.8833) and Jaccard (0.8021), compared with 0.8675 and 0.7773 for nnUNet. It also obtained the lowest mean HD95 (5.4848 mm) and ASSD (1.3839 mm) among the compared methods. Its mean precision was 0.8955, while its recall of 0.8810 was below the highest recall, achieved by DiffUNet (0.9025). These results indicate favorable overall overlap and surface accuracy across the four thoracic OARs.

For the esophagus, CFINet and nnUNet obtained similar DSC values of 0.7754 and 0.7759, respectively. CFINet had lower HD95 (6.1801 versus 7.3248 mm) and ASSD (1.5897 versus 2.2720 mm). For the heart, trachea, and aorta, CFINet achieved the highest DSC among the compared methods. These are organ-at-risk

Table 11 SegTHOR overlap metrics and HD95. Red, blue, and green cells indicate the best, second-best, and third-best scores within each metric and organ, respectively. HD95 is in millimeters.

Metric	Model	Esophagus	Heart	Trachea	Aorta	Mean
DSC	SwinUNet [1]	0.6320	0.8956	0.7217	0.8507	0.7750
	TransUNet [2]	0.7098	0.7750	0.7258	0.8745	0.7713
	UNet++ [24]	0.6666	0.9201	0.7941	0.8951	0.8190
	TransBTS [18]	0.6759	0.9251	0.8288	0.9078	0.8344
	3D UX-Net [11]	0.5882	0.6977	0.8419	0.7097	0.7094
	nnFormer [23]	0.6643	0.9168	0.7677	0.8892	0.8095
	nnUNet [7]	0.7759	0.9335	0.8351	0.9257	0.8675
	SwinUNETR [5]	0.6159	0.9376	0.7841	0.8882	0.8065
	UNETR [6]	0.6439	0.9290	0.8338	0.8763	0.8207
	MedFormer [3]	0.7044	0.9308	0.8432	0.9028	0.8453
	DiffUNet [20]	0.7171	0.8019	0.7555	0.8533	0.7819
	CFINet	0.7754	0.9549	0.8649	0.9379	0.8833
Jaccard	SwinUNet [1]	0.4709	0.8131	0.5716	0.7423	0.6495
	TransUNet [2]	0.5599	0.6373	0.5721	0.7815	0.6377
	UNet++ [24]	0.5198	0.8524	0.6670	0.8128	0.7130
	TransBTS [18]	0.5258	0.8623	0.7151	0.8320	0.7338
	3D UX-Net [11]	0.4269	0.5545	0.7306	0.5593	0.5678
	nnFormer [23]	0.5076	0.8471	0.6283	0.8011	0.6960
	nnUNet [7]	0.6443	0.8774	0.7255	0.8622	0.7773
	SwinUNETR [5]	0.4530	0.8827	0.6538	0.8002	0.6974
	UNETR [6]	0.4839	0.8680	0.7191	0.7812	0.7131
	MedFormer [3]	0.5644	0.8710	0.7327	0.8242	0.7481
	DiffUNet [20]	0.5653	0.6773	0.6120	0.7459	0.6501
	CFINet	0.6430	0.9139	0.7682	0.8834	0.8021
HD95	SwinUNet [1]	25.4773	108.2574	68.3414	37.6240	59.9250
	TransUNet [2]	16.8651	249.1172	94.6251	20.4845	95.2730
	UNet++ [24]	36.1659	29.0544	13.3849	50.8840	32.3723
	TransBTS [18]	8.7543	16.4753	6.7184	10.0629	10.5027
	3D UX-Net [11]	271.8827	309.7362	9.3344	324.4354	228.8472
	nnFormer [23]	11.8887	13.0926	13.8393	8.7376	11.8896
	nnUNet [7]	7.3248	87.9941	40.7821	4.8042	35.2263
	SwinUNETR [5]	17.8697	8.7868	45.4867	6.9938	19.7842
	UNETR [6]	10.5677	17.3809	11.5951	7.9725	11.8790
	MedFormer [3]	44.2319	37.0347	10.7801	49.5722	35.4047
	DiffUNet [20]	22.9106	90.1241	51.0696	88.3662	63.1176
	CFINet	6.1801	5.2044	6.6116	3.9431	5.4848

Table 12 SegTHOR precision, recall, and average symmetric surface distance (ASSD). Red, blue, and green cells indicate the best, second-best, and third-best scores, respectively. ASSD is in millimeters.

Metric	Model	Esophagus	Heart	Trachea	Aorta	Mean
Precision	SwinUNet [1]	0.6134	0.8551	0.6802	0.8250	0.7435
	TransUNet [2]	0.7252	0.6467	0.6279	0.8784	0.7196
	UNet++ [24]	0.7471	0.9061	0.8143	0.9091	0.8441
	TransBTS [18]	0.8530	0.9431	0.8567	0.8784	0.8828
	3D UX-Net [11]	0.5282	0.5652	0.8406	0.5931	0.6318
	nnFormer [23]	0.7349	0.9438	0.7113	0.8867	0.8192
	nnUNet [7]	0.8069	0.9114	0.7876	0.9129	0.8547
	SwinUNETR [5]	0.5689	0.9257	0.8081	0.8675	0.7925
	UNETR [6]	0.6791	0.9254	0.8282	0.8669	0.8249
	MedFormer [3]	0.8002	0.9035	0.8005	0.8632	0.8419
	DiffUNet [20]	0.6520	0.7164	0.6381	0.7770	0.6959
	CFINet	0.8443	0.9570	0.8461	0.9345	0.8955
Recall	SwinUNet [1]	0.6587	0.9418	0.7910	0.8811	0.8182
	TransUNet [2]	0.7046	0.9764	0.8745	0.8794	0.8587
	UNet++ [24]	0.6300	0.9357	0.7877	0.8827	0.8090
	TransBTS [18]	0.5903	0.9095	0.8289	0.9403	0.8172
	3D UX-Net [11]	0.6739	0.9709	0.8618	0.9192	0.8565
	nnFormer [23]	0.6245	0.8949	0.8461	0.8936	0.8148
	nnUNet [7]	0.7566	0.9596	0.9037	0.9396	0.8899
	SwinUNETR [5]	0.6908	0.9513	0.7779	0.9128	0.8332
	UNETR [6]	0.6259	0.9350	0.8561	0.8905	0.8269
	MedFormer [3]	0.6642	0.9613	0.8981	0.9481	0.8679
	DiffUNet [20]	0.8018	0.9246	0.9356	0.9481	0.9025
	CFINet	0.7317	0.9537	0.8970	0.9416	0.8810
ASSD	SwinUNet [1]	5.2363	14.0168	10.8033	5.1165	8.7932
	TransUNet [2]	3.4873	41.6918	12.6776	4.0862	15.4857
	UNet++ [24]	6.6077	5.8743	3.1756	7.5521	5.8024
	TransBTS [18]	2.2304	3.7646	2.1863	2.4745	2.6640
	3D UX-Net [11]	51.7822	87.4976	3.7444	91.8106	58.7087
	nnFormer [23]	2.6104	3.5763	2.7260	2.8405	2.9383
	nnUNet [7]	2.2720	11.0606	7.1771	1.6085	5.5296
	SwinUNETR [5]	5.0478	2.5032	5.9586	2.1886	3.9245
	UNETR [6]	2.9870	3.7961	2.9009	2.3669	3.0127
	MedFormer [3]	6.9388	5.3640	2.0499	6.2895	5.1605
	DiffUNet [20]	4.2675	15.6558	6.1130	9.9903	9.0067
	CFINet	1.5897	1.6097	1.3407	0.9957	1.3839

results on SegTHOR and provide complementary evidence across a different segmentation task. They are not an external evaluation of esophageal tumor GTV segmentation.

Fig. 22 compares mean DSC and FPVF on SegTHOR. CFINet achieved the highest mean DSC (0.8833) and lowest reported mean FPVF (0.051), compared with 0.8675 and 0.090 for nnUNet. The two metrics jointly describe OAR overlap and false-positive volume.

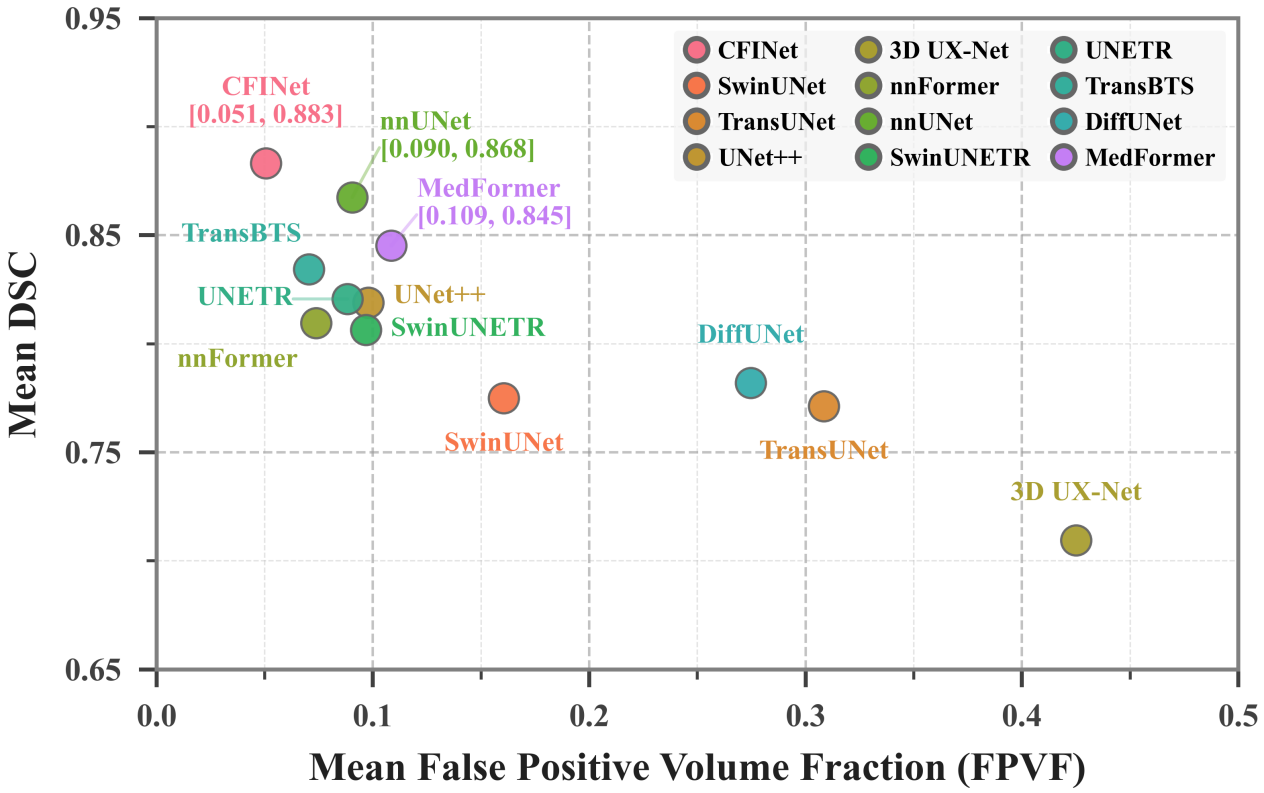

Figure 22 Mean DSC versus false-positive volume fraction (FPVF) for the compared methods on SegTHOR.

Fig. 23 presents the qualitative segmentation results for the four OARs across all compared methods, while Fig. 24 provides enlarged views of the six methods with the highest mean DSC. The enlarged regions show differences between the predicted and reference boundaries. CFINet generally follows the depicted reference contours closely, although these examples do not establish case-level performance for the full dataset.

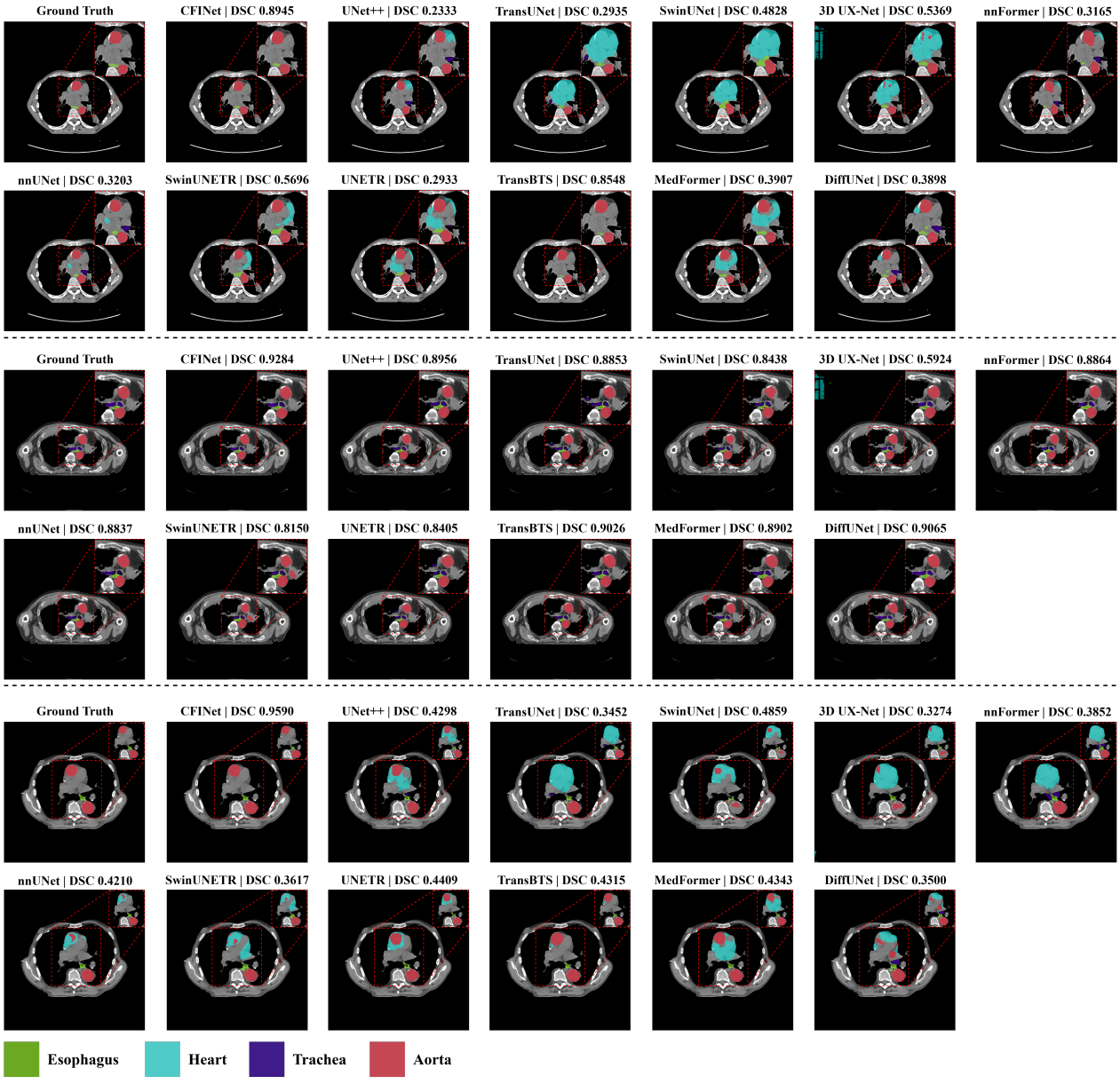

Figure 23 Qualitative comparison of thoracic OAR segmentation on SegTHOR. CT slices and enlarged views show the reference contours and model predictions. The DSC shown above each prediction refers to the depicted example.

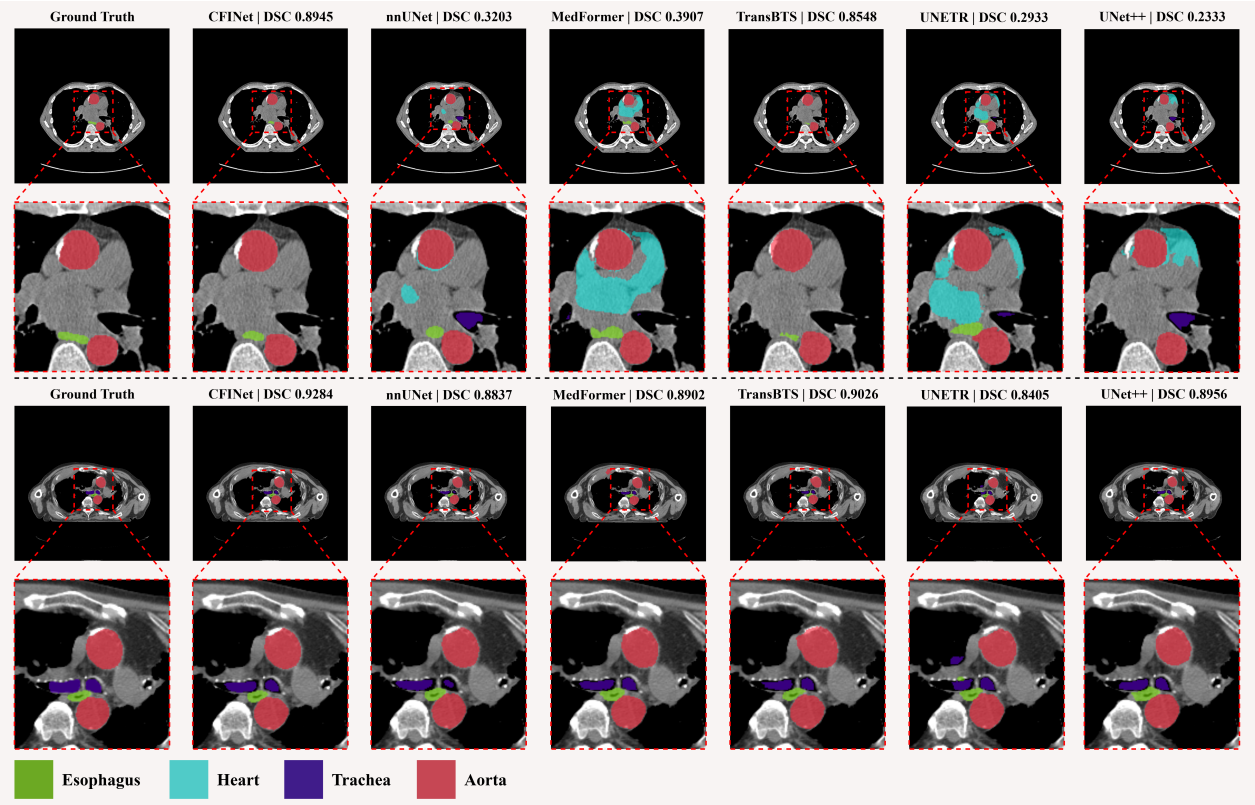

Figure 24 Qualitative comparison of thoracic OAR segmentation on SegTHOR among the six methods with the highest mean DSC. CT slices and enlarged views show the reference contours and model predictions. The DSC shown above each prediction refers to the depicted example.

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